

EDUCATIVE COMMENTARY ON ISI 2016 MATHEMATICS PAPERS

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Introduction

I have been writing educative commentaries on the mathematics papers of the Joint Entrance Examination (JEE) Advanced. Since 2006 the examination became totally MCQ (multiple choice questions) type. For a subject like mathematics where a proof is the soul, this was a severe blow. There are many questions which are not suitable to be asked as MCQ's. Also the paper-setters had to face many other constraints, such as designing questions with a prescribed weightage, rather than proportional to their degrees of difficulty, and, above all, adhering to a uniform pattern common to all three subjects, viz. chemistry, physics and mathematics.

Predictably, the quality of the questions declined. It became difficult for the paper-setters to come up with really new, imaginative questions. Despite this, every year there would be a few really good questions. But their percentage declined alarmingly. Suggestions were made to go back to the old system of a screening round based entirely on MCQ's and the advanced round (where the number of candidates would be reduced as to permit manual evaluation) entirely on full length questions. But they fell on deaf ears.

A ray of hope came with the entrance examinations of ISI (Indian Statistical Institute, Kolkata) for its B.Sc. programmes in Mathematics/Statistics. They have two papers, one entirely MCQ type and the other entirely conventional type where a candidate has to give justifications. While it is hard to

make a generalisation from a single year, if the papers of 2016 are anything to go by, the first paper contains 30 well chosen MCQ's. Two of these were cited with approbation in the JEE commentary 2017 in its concluding remarks. The idea was to point out that even with the constraints of an MCQ format, there can be well designed questions.

After the JEE 2017 commentary was put on the blog, a few readers have requested me to write a full commentary on the ISI papers. My intention was merely to cite examples so that the JEE organisers can take cue and take steps to redeem the original glory of JEE. It was not my intention to write a full commentary on ISI papers. But a few readers have expressed that unlike the JEE questions, where keys and solutions are easily available from a number of coaching institutes, it is hard to get the solutions of the ISI problems. All that is published by ISI is the key (but not the solutions) to the MCQ's in the first paper. This is a sad commentary on the heavy commercialisation of JEE. It also means that even though the ISI problems are better sources to learn from, the average students have no chance to learn unless guided by some mentors.

As an experiment, here is an educative commentary on both the ISI Entrance Examination for B.Math/B.Stat papers (coded respectively as UGA and UGB). Comments are solicited as to whether it is a good source of learning for the students (which is, after all, the avowed purpose of the commentaries on the JEE). Depending on the response I shall prepare the commentaries for other years. The feedback may be sent to me on kd-joshi314@gmail.com or by an sms or a WhatsApp message on my mobile 9819961036.

It is to be noted that all the thirty questions in Paper 1 are MCQ's with only one correct answer. This sometimes provides an unwarranted short cut to the solution, of which I have often been critical in my JEE commentaries. The ISI paper setters must be aware of this. But such questions were included possibly on two grounds, one, sometimes even realising that such a short cut is possible indicates some alertness on the part of a candidate and secondly, and more importantly, the final selection is not based entirely on MCQ's as is the case with JEE.

As this commentary is an offshoot of the educative commentary on JEE, as in the parent commentary, all the references in the present commentary will be to *Educative JEE Mathematics* by the author, unless otherwise stated. The pagination will be as per its third edition, which is available in the market. Also from time to time, comparison will be made with JEE which

is a far more well known examination.

In passing, I have had a long, first hand experience of JEE work in its various facets. I willingly disassociated myself from it from 2003 on ethical grounds when my book *Educative JEE Mathematics* was published. Even after that I was in the service of IIT Bombay till 2013. And even after retirement, I have been writing a commentary every year. I am happy to say that even though I have criticised the JEE paper setters many times, such a criticism has never been taken personally.

With ISI I have no first hand or even a remote association. Although this commentary is motivated by an approbation of the ISI entrance test, in dealing with individual questions, I shall sometimes be advancing my own opinions and comments. I hope that if the ISI organisers come to know of any critical remarks I may make, they will take them in the right spirit (or simply ignore them, if they prefer!).

ISI BStat-BMath-UGA-2016 Paper with Comments

N.B.: In every question only ONE of the given options is correct.

Q.1 The largest integer n for which $n + 5$ divides $n^5 + 5$ is

- (A) 3115. (B) 3120. (C) 3125. (D) 3130.

Answer and Comments: (A). As the problem deals with divisibility by $n + 5$ rather than by n , it is better to introduce a single symbol, say, u for $n + 5$. Then $n = u - 5$ and so the problem reduces to finding the largest integer u which divides $(u - 5)^5 + 5$. On expansion, this is a sum of some powers of u and the constant term $-5^5 + 5 = -3120$. All the powers of u are divisible by u and so the problem really asks to find the largest divisor of the number 3120. Obviously this is 3120 itself. Taking $u = 3120$, n comes out as 3115. Hence (A) is the correct choice.

This is a good problem which tests the ability to focus on what is relevant. The problem unfolds itself the moment you introduce $u = n + 5$. The binomial theorem is not needed in its full strength. Only the constant term in the expansion is relevant. The only arithmetic needed is to calculate 5^5 . Powers of single digit numbers upto the cube are popular with those who love numbers. Some people have a fascination for such powers. Many people remember the squares of the integers from 1 to 30. A few persons remember the cubes of some smaller numbers. (A well known anecdote about the great mathematician **Ramanujan** is that when somebody mentioned to him the number 1729 as not a particularly interesting number, he instantly quipped that it is the smallest integer which is the sum of two perfect cubes in two ways. He was right, for $1729 = 1000 + 729 = (10)^3 + (9)^3$ and also $1729 = 1728 + 1 = (12)^3 + (1)^3$.) For higher powers, the memory is predictably more limited. But once in a while it pays, although not so much in this particular problem. It would, in fact, be a dicey problem if the data involves a huge number, say 69343957, and the success depends crucially on recognising it as a perfect power (in the present case $(37)^5$).

Powers of 2 are important in combinatorics and hence also in computer science. The fact that $2^{10} = 1024$ is only slightly bigger than

$(10)^3$ is convenient in getting a rough idea how big 2^n is for a large n .

However, in the present problem the success does not depend on recognising 3125 as 5^5 right at the start. Those who know this will only save a few seconds in calculating 5^5 . By giving all the options closely related to 5^5 , the paper setters have given a built in warning lest a good student commit a computational mistake at the end. So this problem meets all the requirements of a good question. Once the key idea strikes, the further work is not laborious. Moreover, it protects a good student and does not allow any sneaky path.

Q.2 Let p, q be primes and a, b be integers. If pa is divided by q , the remainder is 1. If qb is divided by p , then also the remainder is 1. The remainder when $pa + qb$ is divided by pq is

- (A) 1. (B) 0. (C) -1 . (D) 2.

Answer and Comments: (A). It is ironical that after commending the last problem as a good one in all respects, the very next problem should invite some criticism. All the four options given are fixed numbers, independent of a particular choice of the primes p and q . Moreover, only one of them is the correct answer. So, the correct answer can be identified by taking the primes p, q of our choice. The simplest case is to take $p = 2$ and $q = 3$. We want $2a$ to leave a remainder when divided by 3. The simplest choice is to take $a = 2$. Similarly, $3b$ is to be an odd number and so b can be taken to be any odd number. Why not, then $b = 1$? With these choices, $pa + qb = 4 + 3 = 7$ which leaves a remainder 1 when divided by pq which is 6.

No further work needs to be done. A good candidate might also try some other pair of primes, say $p = 3$ and $q = 5$ to confirm the answer. In that case we can take $a = 2$ and $b = 2$ too. That gives $pa + qb = 6 + 10 = 16$ which leaves a remainder 1 when divided by pq which is 15.

It is hard to believe that the paper setters missed this sneaky answer. It is probably because in the present problem, an honest approach is easy too. Indeed, if we write $pa = qm + 1$ and $qb = pn + 1$ for some integers m and n , we can write $pa + qb$ as $pa + pn + 1$, which means that the integer $pa + qb - 1$ is divisible by p . But we can also write

$pa + qb$ as $qm + 1 + qb$ and see that $pa + qb - 1$ is divisible by q too. Since $pa + qb - 1$ is divisible by both the primes p and q , it is divisible by the product pq . (Here we are tacitly assuming that p, q are distinct. This is to be inferred from the notations. Scrupulous paper setters would generally specify that p, q are distinct. But then they might also invite the criticism of being pedantic. This is probably what dissuaded them.) Whatever be the case, the present problem is not suitable as an MCQ where it is also given that only one answer is correct.

The sneak path could have been prevented by giving in each options, two possible choices, say (A) as ‘either 1 or 0’, (B) as ‘either 1 or -1’ and so on. Such brainwork is common with the JEE paper setters. The ISI paper setters are yet to learn! Or probably, they stay assured that they have the cushion of Paper 2 where the candidates have to give reasoning and therefore have no qualms about allowing the speedy to score in Paper 1.

Q.3 The polynomial $x^7 + x^2 + 1$ is divisible by

- (A) $x^5 - x^4 + x^2 - x + 1$. (B) $x^5 + x^4 + 1$.
 (C) $x^5 + x^4 + x^2 + x + 1$. (D) $x^5 - x^4 + x^2 + x + 1$.

Answer and Comments: (A). A brute force approach would be to take each of the options and apply long division to see if the polynomial in it is a divisor of $x^7 + x^2 + 1$. Although not recommended, a possible advantage of this approach is that once you find a polynomial that is a divisor, you need not try the remaining options as it is given that only one of the answers is correct.

An intelligent approach is to note that all the given four polynomials are of degree 5. Moreover their leading coefficients and the constant terms are 1 in all cases. As this is also the case with $x^7 + x^2 + 1$, whenever the polynomial in any option divides $x^7 + x^2 + 1$, the other factor must be of the form $x^2 + ax + 1$ for some constant a . This gives us an equation

$$f(x)(x^2 + ax + 1) = x^7 + x^2 + 1 \quad (1)$$

where $f(x)$ stands for one of the four given polynomials. We can simplify this a little by subtracting $x^2 + ax + 1$ from both the sides to get

$$g(x)(x^2 + ax + 1) = x^7 - ax = x(x^6 - a) \quad (2)$$

where $g(x) = f(x) - 1$. $g(x)$ is a monic polynomial (i.e. a polynomial with leading coefficient 1) with constant term 0 and so has x as a factor. Cancelling it we get

$$h(x)(x^2 + ax + 1) = x^6 - a \quad (3)$$

where $h(x)$ is a monic polynomial of degree 4. Note that in none of the four options, there is an x^3 term. So in $h(x)$ there cannot be an x^2 term. Hence $h(x)$ is of the form

$$h(x) = x^4 + b_1x^3 + b_2x + b_3 \quad (4)$$

for some constants b_1, b_2, b_3 . Putting this into (3) and equating the constant terms of both the sides gives $b_3 = -a$. Also, equating the coefficients of x^5 of both the sides gives $b_1 + a = 0$, i.e. $b_1 = -a$. We have not yet determined b_2 and a . But from the information already obtained and the fact that $f(x) = g(x) + 1 = xh(x) + 1$ we see that for the correct option $f(x)$ must be of the form

$$f(x) = x^5 - ax^4 + b_2x^2 - ax + 1 \quad (5)$$

This rules out (B) and (D) because the coefficients of x^4 and x are not equal in them. To see which of (A) and (C) is correct, we investigate further. In the simplified form, $h(x) = x^4 - ax^3 + b_2x - a$ and so from (3) we get

$$(x^4 - ax^3 + b_2x - a)(x^2 + ax + 1) = x^6 - a \quad (6)$$

Equating the coefficients of x^6, x^5 and x^0 (i.e. the constant terms) from the two sides is not going to give us anything new now. But if we equate the coefficients of x^4 we get $1 - a^2 = 0$ and hence $a = \pm 1$. Also from the coefficients of x we get $b_2 = a^2 = 1$. We are still unable to distinguish between (A) and (C). So we now equate the coefficients of x^3 on both the sides to get $-a + b_2 = 0$. As we know $b_2 = 1$, we finally get $a = 1$. We now see from (5) that (A) is the only correct option.

Instead of determining a, b_1, b_2, b_3 in this piecemeal manner we could have, after putting (4) into (3), equated the coefficients of all like powers on both the sides to get a system of seven equations in the four unknowns a, b_1, b_2, b_3 . Some of these equations would be redundant or repetitious. But solving them we would get $a = 1, b_1 = b_3 = -1$ and $b_2 = 1$. Hence (A) is correct. But sometimes it pays to do only halfway work and pause to see if by chance the partial work is enough to give the answer. (This would have been the case if in the option (C) the coefficients of x^4 and x were unequal.)

There is a shorter, albeit trickier method to tackle the problem. It is based on ω , the complex cube root of unity. Problems involving ω are fairly common in the JEE papers, the most recent examples being Q.59 (C) of Paper 1 of JEE 2015 and Q. 54 of Paper 1 of JEE 2016. As commented in the latter, most of the problems involving ω can be tackled using two basic properties, (i) its powers recur in a cycle of length 3, that is, $\omega^m = \omega^n$ whenever m, n are integers that differ by a multiple of 3 and (ii) $\omega^2 + \omega + 1 = 0$, and hence $x^2 + x + 1$ factors as $(x - \omega)(x - \omega^2)$. Note also that $\omega^2 = \bar{\omega}$. As a result, problems which refer to ω overtly are relatively easy. Those which make only an indirect reference to it through the polynomial $x^2 + x + 1$ are rather tricky. See, for example, Exercise (7.15). But even these have been asked before. (See Comment No.12 of Chapter 7.)

In our problem at hand, the reference to ω is even more covert. ω is a root of $x^2 + x + 1 = 0$. But because of Property (i) above, we can replace the x term by x^7 (but retain x^2 as it is). Then we get that ω is also a root of $x^7 + x^2 + 1$, which is exactly the polynomial whose divisors we are looking for. Of course, as a polynomial of degree seven it may have many other roots. But as long as ω is a root, so is $\bar{\omega}$ since the polynomial has real coefficients. As a result, we know that $(x - \omega)(x - \bar{\omega})$ i.e. $x^2 + x + 1$ is a factor of the given polynomial $x^7 + x + 1$. The other factor (which will have degree 5) can be found by long division. But even here there is a short cut. Suppose this other factor is $f(x)$. Then

$$f(x)(x^2 + x + 1) = x^7 + x^2 + 1 \quad (7)$$

Subtracting $x^2 + x + 1$ from both the sides and letting $g(x) = f(x) - 1$,

we get

$$g(x)(x^2 + x + 1) = x^7 - x = x(x^6 - 1) \quad (8)$$

These are similar to (1) and (2) respectively. But now the advantage is that we know that $a = 1$. Factorising $x^6 - 1$ as $(x^3 - 1)(x^3 + 1)$ and further as $(x - 1)(x^2 + x + 1)(x^3 - 1)$ we see that $g(x) = x(x - 1)(x^3 + 1) = x^5 - x^4 + x^2 - x + 1$ and hence $f(x) = x^5 - x^4 + x^2 - x + 1$. So (A) is the correct option.

Yet another possibility is, of course, to do some guesswork. The R.H.S. of (3) is a difference of two sixth powers no matter what a is. And in case a happens to be 1, then it factors as $(x^3 + 1)(x^3 - 1)$. $x^2 + x + 1$ is a factor of $x^3 - 1$. The other factor, viz. $x - 1$ multiplied by $x^3 + 1$ (the other factor of $x^6 - 1$) will give $h(x)$ as $x^4 - x^3 + x - 1$. And that means $f(x) = xh(x) + 1 = x^5 - x^4 + x^2 - x + 1$.

Although the problem is elementary, the work needed in arriving at the solution is a bit longer than would be justified in a short question unless you are able to hit the jackpot through ω . Guesswork pays in such problems, if applied after reducing a problem to a more manageable form which would permit such guesswork. In the present problem, getting (3) is this vital step. So the problem may be said to have been designed to reward intelligent guesswork.

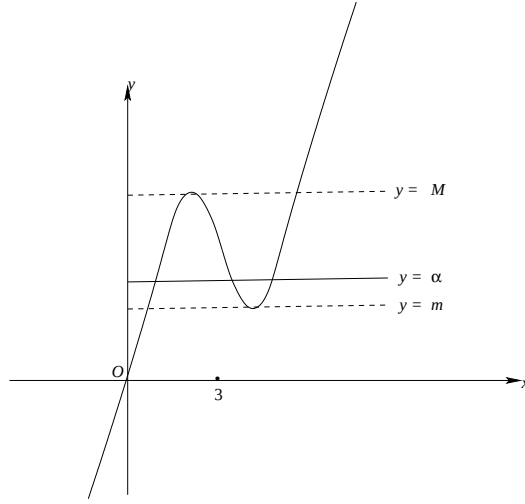
Q.4 Let $\alpha > 0$. If the equation $p(x) = x^3 - 9x^2 + 26x - \alpha$ has three positive real roots, then α must satisfy

- (A) $\alpha \leq 27$. (B) $\alpha > 81$.
 (C) $27 < \alpha \leq 54$. (D) $54 < \alpha \leq 81$.

Answer and Comments: (A). The roots of $x^3 - 9x^2 + 26x - \alpha$ correspond to the points of intersection (or, rather, their x -coordinates) of the graph $y = x^3 - 9x^2 + 26x$ with the horizontal line $y = \alpha$. So the problem can be paraphrased to ask for which values of α , the line $y = \alpha$ cuts the graph of $y = f(x) = x^3 - 9x^2 + 26x$ in three points on the right of the y -axis.

In general the graph of a cubic function $f(x)$ (with a positive leading coefficient) is either strictly increasing throughout or is increasing till it

encounters a local maximum, say M , then decreasing till it encounters a local minimum, say m and then increasing again. In the first case, every horizontal line cuts the graph at precisely one point. In the second case the number of points of intersection of $y = f(x)$ with the line $y = \alpha$ is 1 for $\alpha < m$ and $\alpha > M$, 2 if $\alpha = m$ or $\alpha = M$ and 3 if $m < \alpha < M$. (See the figure below.)



So first we determine the local maximum and the local minimum of the cubic $y = f(x) = x^3 - 9x^2 + 26x$. Differentiating, $f'(x) = 3x^2 - 18x + 26$. This has two real roots, viz., $\frac{9 \pm \sqrt{3}}{3} = 3 \pm \frac{1}{\sqrt{3}}$. So the function has a local maximum at $x = 3 - \frac{1}{\sqrt{3}}$, a local minimum at $x = 3 + \frac{1}{\sqrt{3}}$ and a point of inflection at $x = 3$. The line $y = \alpha$ will cut the graph in three distinct points if and only if α lies between the local minimum $f\left(3 + \frac{1}{\sqrt{3}}\right)$ and the local maximum $f\left(3 - \frac{1}{\sqrt{3}}\right)$. By a direct calculation,

$$f\left(3 + \frac{1}{\sqrt{3}}\right) = \left(3 + \frac{1}{\sqrt{3}}\right)^3 - 9\left(3 + \frac{1}{\sqrt{3}}\right)^2 + 26\left(3 + \frac{1}{\sqrt{3}}\right)$$

Expanding the cube and the square and collecting together the rational

and the irrational terms, this becomes

$$\begin{aligned}
f\left(3 + \frac{1}{\sqrt{3}}\right) &= 27 + \frac{27}{\sqrt{3}} + \frac{9}{3} + \frac{1}{3\sqrt{3}} \\
&\quad - 9\left(9 + \frac{1}{3} + \frac{6}{\sqrt{3}}\right) + 78 + \frac{26}{\sqrt{3}} \\
&= (27 + 3 - 81 - 3 + 78) + \frac{1}{\sqrt{3}}\left(27 + \frac{1}{3} - 54 + 26\right) \\
&= 24 - \frac{2}{3\sqrt{3}}
\end{aligned} \tag{1}$$

The calculation of the local maximum $f\left(3 - \frac{1}{\sqrt{3}}\right)$ is similar.

$$\begin{aligned}
f\left(3 - \frac{1}{\sqrt{3}}\right) &= \left(3 - \frac{1}{\sqrt{3}}\right)^3 - 9\left(3 - \frac{1}{\sqrt{3}}\right)^2 + 26\left(3 - \frac{1}{\sqrt{3}}\right) \\
&= 27 - \frac{27}{\sqrt{3}} + \frac{9}{3} - \frac{1}{3\sqrt{3}} \\
&\quad - 9\left(9 + \frac{1}{3} - \frac{6}{\sqrt{3}}\right) + 78 - \frac{26}{\sqrt{3}} \\
&= (27 + 3 - 81 - 3 + 78) - \frac{1}{\sqrt{3}}\left(27 + \frac{1}{3} - 54 + 26\right) \\
&= 24 + \frac{2}{3\sqrt{3}}
\end{aligned} \tag{2}$$

(We could also write this directly from (1) by observing that in the binomial expansions, the rational terms are the same but the irrational terms, which come from odd powers of $\frac{1}{\sqrt{3}}$ are multiplied by -1 .)

So the equation $p(x) = x^3 - 9x^2 + 26x - \alpha$ will have three real roots if and only if $24 - \frac{2}{3\sqrt{3}} < \alpha < 24 + \frac{2}{3\sqrt{3}}$. It is important to note that the lower bound $24 - \frac{2}{3\sqrt{3}}$ is positive. Moreover, the point where $f(x)$ has a local maximum, viz. $3 - \frac{1}{\sqrt{3}}$ is also positive. That ensures that all the three points of intersection of $y = f(x)$ and $y = \alpha$ lie on the right of the y -axis, since $f(x) < 0$ for $x < 0$.

So the exact range of α is the interval $(24 - \frac{2}{3\sqrt{3}}, 24 + \frac{2}{3\sqrt{3}})$. (We can include the end-points too if we do not require the three roots to be distinct.) Since $0 < \frac{2}{3\sqrt{3}} < 1$, it follows that $\alpha \in (23, 25)$. So (A) is correct.

The key idea in the solution is to correlate the roots of the polynomial $p(x) = x^3 - 9x^2 + 26x - \alpha$ with the points of intersection of the graph of $y = x^3 - 9x^2 + 26$ with the line $y = \alpha$. Once this strikes, the calculation of the local maximum and the local minimum is a very routine matter. It is not clear what is gained by making the correct option, viz. (A) weaker than what is actually true. Perhaps the idea was to avoid irrational numbers, which might give an unwarranted hint to the candidate. A more likely possibility is that the paper setters had some other method in mind which is good enough to imply only (A). One such method is based on the A.M.-G.M. inequality. Let a, b, c be the three roots of $p(x) = x^3 - 9x^2 + 26x - \alpha$. Then $p(x)$ factors as $(x - a)(x - b)(x - c)$ and equating the coefficients of the like powers we get three well known equations, viz.

$$a + b + c = 9 \quad (3)$$

$$ab + bc + ca = 26 \quad (4)$$

$$\text{and } abc = \alpha \quad (5)$$

From (3), (5) and the A.M.-G.M. inequality, we get

$$3 = \frac{a + b + c}{3} \geq (abc)^{1/3} = \alpha^{1/3} \quad (6)$$

whence $\alpha \leq 27$. Actually, we get strict inequality because for equality to hold in (6), we must have $a = b = c = 3$. But in that case (4) fails.

If instead of (3) and (5) we take (4) and (5) and apply the A.M.-G.M. inequality, we get

$$\frac{26}{3} = \frac{ab + bc + ca}{3} \geq (a^2b^2c^2)^{1/3} = \alpha^{2/3} \quad (7)$$

whence $\alpha^2 \leq (\frac{26}{3})^3$. This gives $\alpha \leq (\frac{26}{3})^{3/2}$. This is a better upper bound than 27 because $(\frac{26}{3})^{3/2} < (\frac{27}{3})^{3/2} = 27$. But it is weaker than the upper bound $24 + \frac{2}{3\sqrt{3}}$ which we got by the calculus method.

It is instructive to analyse why we get different upper bounds on α depending upon which method we use. If a, b, c are all positive then it is easy to see why in general (7) provides a better (i.e. a lower) upper bound on $\alpha = abc$ than (6) does. The inequality we get from (6) is

$$\alpha = abc \leq \left(\frac{a+b+c}{3}\right)^3 \quad (8)$$

while from (7) we get

$$\alpha = abc \leq \left(\frac{ab+bc+ca}{3}\right)^{3/2} \quad (9)$$

We claim that the R.H.S. of (8), say x , is in general greater than or equal to that of (9), say y . As all the numbers involved are positive, we might as well compare $x^{2/3}$ with $y^{2/3}$, i.e. $\left(\frac{a+b+c}{3}\right)^2$ with $\frac{ab+bc+ca}{3}$. This reduces to comparing $(a+b+c)^2$ with $3(ab+bc+ca)$. From the identity

$$a^2 + b^2 + c^2 - (ab + bc + ca) = \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2] \quad (10)$$

we see that

$$a^2 + b^2 + c^2 \geq ab + bc + ca \quad (11)$$

because the R.H.S. of (10) is positive except when $a = b = c$. Adding $2(ab + bc + ca)$ to both the sides gives the desired inequality, viz. that the R.H.S. of (9) is at most that of (8).

Let us now see why the calculus upper bound is superior to both these upper bounds. We may say that this happens because calculus, equipped with the power of the limiting process is a more evolved branch of mathematics than algebra. But that is a very generalistic answer. To really nab the reason, let us go back to the figure we drew at the start of the solution, viz. the graph of a cubic. There we saw that the line $y = \alpha$ cuts the graph at points whose x -coordinates are the roots of the polynomial $p(x) = x^3 - 9x^2 + 26x - \alpha$. For $m < \alpha < M$ these three points are distinct. The A.M.-G.M. inequality for three numbers is generally not sharp unless the three numbers are equal. There is a lot of waste in its proof even in the simplest case of two numbers. For

example, if $a \neq b$, then $a^2 + b^2 > 2ab$ holds because $(a - b)$ is non-zero and hence has a positive square. But in general $a - b$ could be a numerically large number and merely saying that its square is positive is an understatement. If the three numbers are widely spread out then neither (8) nor (9) gives a good upper bound on their product. Take, for example, the numbers 1, 8 and 27. Their product is 216. But the upper bound given by (8) is $(\frac{1+8+27}{3})^3 = (12)^3 = 1728$. The upper bound given by (9) is $(\frac{8+27+216}{3})^{3/2} = (\frac{251}{3})^{3/2}$ which is slightly bigger than $(\frac{243}{3})^{3/2} = (81)^{3/2} = 9^3 = 729$. Although much better than the earlier upper bound 1728, it is still way far off from the actual value of α which is 216. (Try what happens with close numbers like 3, 4 and 5.)

Now, for a given α , let a, b, c be the roots of $p(x) = x^3 - 9x^2 + 26x - \alpha$. Note that they can never be equal, because if they were then by (3) we have $a = b = c = 3$ which is not consistent with (4). This explains why the A.M.-G.M. inequality, no matter how cleverly applied cannot provide the best upper bound, viz. $24 + \frac{2}{3\sqrt{3}}$ on α which we got by the calculus method. When three numbers a, b, c cannot all be equal, the best that can happen is when two of them are equal. This is what happens for $\alpha = M$ in the figure above. For, the line $y = M$ cuts the graph of $y = x^3 - 9x^2 + 26x$ into two coinciding points (i.e. the touches it). Let a be the x -coordinate of this point of contact. The third point of intersection, has a different x -coordinate, say b , which corresponds to the third root of $p(x)$. So, for $\alpha = M$, the three roots of $p(x)$ are a, a and b with $b > a$. (For $\alpha = m$, the three roots are a, b, b with $a < b$.)

So, the upper bound M on α is the value of a^2b (the product of the roots a, a, b). These roots are given by (3) and (4) which now reduce to

$$2a + b = 9 \quad \text{and} \quad a^2 + 2ab = 26 \quad (12)$$

This is a system of two equations in two unknowns. It is easy to solve it. Substituting $b = 9 - 2a$ into the second equation gives a quadratic $3a^2 - 18a + 26 = 0$. Its roots are $3 \pm \frac{1}{\sqrt{3}}$. As we are interested in an upper bound rather than a lower bound on α , we take the root $3 - \frac{1}{\sqrt{3}}$ and call it a . The value of b then comes out to be $9 - 6 + \frac{2}{\sqrt{3}} = 3 + \frac{2}{\sqrt{3}}$.

By a direct calculation, $\alpha = a^2b$ comes out as

$$\begin{aligned}
\alpha = a^2b &= \left(3 - \frac{1}{\sqrt{3}}\right)^2 \left(3 + \frac{2}{\sqrt{3}}\right) \\
&= \left(\frac{28}{3} - \frac{6}{\sqrt{3}}\right) \left(3 + \frac{2}{\sqrt{3}}\right) \\
&= 28 - 4 + \frac{56}{3\sqrt{3}} - \frac{18}{\sqrt{3}} = 24 + \frac{2}{\sqrt{3}}
\end{aligned} \tag{13}$$

which is exactly the same answer we got earlier by calculating $f(a)$ where $f(x) = x^3 - 9x^2 + 26x$.

So now we see why the calculus method gives the best upper bound on α . It applies the A.M.-G.M. inequality in the second least wasteful situation where two of the three numbers are equal.

There is yet another way to see why the calculus method works better. The problem involves finding the maximum of $\alpha = abc$ subject to the two equations (3) and (4). Here α is a function of three real variables a, b, c . But these three variables are not free to take any values. The equations (3) and (4) are examples of what are called **constraints** on a, b, c . So this problem can be looked upon as problem of constrained optimisation. Problems of trigonometric optimisation also fall under constrained optimisation, the constraint being that the three angles of a triangle add to 180 degrees. There is a well-known method, based on what are called **Lagrange's multipliers** for constrained optimisation when there is only one constraint. It is beyond the level of HSC, but is illustrated with an example in Comment No. 4 of Chapter 14. The case of two or more constraints is more complicated, because the maximum subject to one constraint may occur at a point which does not satisfy the second constraint. That is exactly what happens here. We have to look for the maximum among the points (a, b, c) which satisfy *both* (3) and (4) simultaneously. One way out is to use these constraints to express α as a function of only one variable and then maximise it using calculus. This can be done by writing

$$\begin{aligned}
\alpha = abc &= a(bc) = a(26 - ab - ac) \\
&= a(26 - a(b + c)) = a(26 - a(9 - a)) \\
&= a^3 - 9a^2 + 26a
\end{aligned} \tag{14}$$

which is exactly the same cubic we started with. So we are back to square one. In fact the danger now is to think that α has no maximum because $a^3 - 9a^2 + 26 \rightarrow \infty$ as $a \rightarrow \infty$. The catch, of course, is that the requirement that a, b, c be all positive (and, in particular real) rules out unconditional variation of a . Determining the permissible variation of a and maximising the cubic on this range will be a more difficult job than any of the methods we have used so far.

The stipulation that $\alpha > 0$ in the statement is unnecessary in the present problem if attempted by the calculus method. It would have been relevant if the point of local minimum lay below the x -axis. Also if the point of the local maximum on the graph lay on the left of the x -axis, then we would have to restrict α to the interval $(m, f(0))$. With the given data these complications do not arise. But that leaves open the question whether a candidate who answers the question correctly has really applied his/her mind to address these subtle points or whether ignorance is bliss for him/her. This, of course, is the fate of many MCQ's and again, is excusable when there is another paper where there is room to distinguish the scrupulous from the unscrupulous. As the question stands, both the stipulation and the requirement that all the three roots of $x^3 - 9x^2 + 26x - \alpha$ are positive are redundant for the calculus method. But they are needed for the algebraic method since the A.M.-G.M. inequality presupposes that all the numbers involved are positive. And this is probably all that the paper setters intended to test.

Q.5 The largest integer which is less than or equal to $(2 + \sqrt{3})^4$ is

- (A) 192. (B) 193. (C) 194. (D) 195.

Answer and Comments: (B). The problem is supposed to be done without calculators. A direct expansion of the number $(2 + \sqrt{3})^4$ gives

$$(2 + \sqrt{3})^4 = ((2 + \sqrt{3})^2)^2 = (7 + 4\sqrt{3})^2 = 97 + 56\sqrt{3} \quad (1)$$

So the problem reduces to determining which of the numbers 95, 96, 97 and 98 (obtained by subtracting 97 from the given four options) equals the integral part of $56\sqrt{3}$. This is best done by directly comparing squares. Because 56 and 98 have a large common factor, viz. 14, it is easiest to compare $56\sqrt{3}$ with 98. Cancelling 14 and squaring we get

$(4\sqrt{3})^2 = 48$ on one side and $7^2 = 49$ on the other. So $56\sqrt{3} < 98$ and hence the given number is less than 195.

The next easier number to compare $56\sqrt{3}$ with is 96 because of the common factor 8. Since $(7\sqrt{3})^2 = 147$ while $12^2 = 144$, we see that $56\sqrt{3} > 96$. That takes us to the most delicate comparison, viz. that between $56\sqrt{3}$ and 97. Now there are no common factors. Fortunately, the squares are easy to calculate without the brute force method. $(56)^2 = (50 + 6)^2 = 2500 + 600 + 36 = 3136$, while $97^2 = (100 - 3)^2 = 10000 - 600 + 9 = 9409$. Thus $(56\sqrt{3})^2 = 3136 \times 3 = 9408$ is only slightly less than $(97)^2$. So the integral part of $56\sqrt{3}$ is 96 and hence that of $(2 + \sqrt{3})^4$ is $97 + 96 = 193$.

There is a more elegant solution based on the concept of the **conjugate surd** of $2 + \sqrt{3}$ defined as $2 - \sqrt{3}$. Analogously to (2), we have

$$(2 - \sqrt{3})^4 = ((2 - \sqrt{3})^2)^2 = (7 - 4\sqrt{3})^2 = 97 - 56\sqrt{3} \quad (2)$$

Hence

$$(2 + \sqrt{3})^4 + (2 - \sqrt{3})^4 = 194 \quad (3)$$

(The reasoning applied here is the same as that in the alternate derivation of (2) in the solution to the last question. The irrational terms in the binomial expansions of the two surds cancel out.)

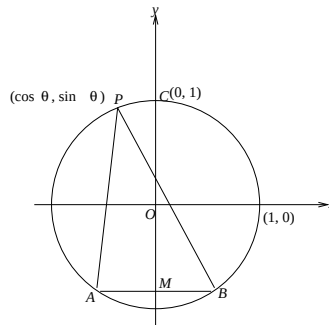
Now the number $2 - \sqrt{3} \in (0, 1)$. So all its powers lie in $(0, 1)$. Hence the second term of (4) is some number between 0 and 1. Therefore the integral part of the first term is $194 - 1 = 193$.

This second solution is akin to the solution of a 1988 JEE problem, which asks to show that if $R = (5\sqrt{5} + 11)^{2n+1}$ and $f = R - [R]$, then $Rf = 4^{2n+1}$. (See Exercise (6.27)(a).)

Q.6 Consider a circle of unit radius and a chord of that circle that has unit length. The area of the largest triangle that can be inscribed in that circle with that chord as the base is

- | | |
|--|--|
| (A) $\frac{1}{2} + \frac{\sqrt{2}}{4}$ | (B) $\frac{1}{2} + \frac{\sqrt{2}}{2}$ |
| (C) $\frac{1}{2} + \frac{\sqrt{3}}{4}$ | (D) $\frac{1}{2} + \frac{\sqrt{3}}{2}$ |

Answer and Comments: (C). A very straightforward problem on maximisation. But since the data as well as the function to be maximised (viz. the area of a certain triangle) are geometric, the problem comes under what is called geometric optimisation. For such problems, instead of the general method of calculus, certain simple facts from geometry (e.g. that the shortest path joining two points is the straight line segment) can provide a much simpler solution. (For an illustration see the problem at the end of Comment No. 15 in Chapter 14.)



Let O be the centre of the circle and AB be the given chord. Let P be the third vertex of a triangle inscribed in the circle with base AB . Then the area of $\triangle PAB$ will be maximum when P is farthest away from AB . That will happen when P is at C , the farther end of the diameter passing through the midpoint M of AB . When this happens, $CM = CO + OM = 1 + \sqrt{1 - \frac{1}{4}} = 1 + \frac{\sqrt{3}}{2}$ and the area is $CM \cdot MB = \frac{1}{2} + \frac{\sqrt{3}}{4}$.

For a calculus based solution, we need to express the position of the variable point P in terms of a single variable. One possible choice is to introduce coordinates so that the circle is $x^2 + y^2 = 1$, $A = (-\frac{1}{2}, \frac{\sqrt{3}}{2})$ and $B = (\frac{1}{2}, \frac{\sqrt{3}}{2})$. Then P can be taken as $(\cos \theta, \sin \theta)$ for some $\theta \in [0, 2\pi]$. The area of $\triangle PAB$ is $f(\theta)$ where

$$f(\theta) = \frac{1}{2}(\sin \theta + \frac{\sqrt{3}}{2}) \quad (1)$$

We hardly need derivatives to maximise $f(\theta)$. It is maximum when $\sin \theta$ is maximum. That happens when $\theta = \frac{\pi}{2}$. The maximum value of $f(\theta)$ is $\frac{1}{2} + \frac{\sqrt{3}}{4}$.

Either way, the problem is extremely simple and not comparable to any of the earlier questions either in terms of originality or in terms of degree of difficulty. Such problems, which even a mediocre student can

solve with essentially the same amount of work are effectively useless in a competitive selection.

Q.7 Let $z_1 = 3 + 4i$. If z_2 is a complex number such that $|z_2| = 2$, then the greatest and the least values of $|z_1 - z_2|$ are respectively

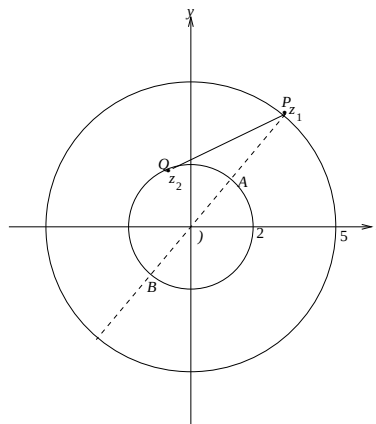
(A) 7 and 3.

(B) 5 and 1.

(C) 9 and 5.

(D) $4 + \sqrt{7}$ and $\sqrt{7}$.

Answer and Comments: (A). Another problem on geometric optimisation except that ostensibly, the data is in terms of complex numbers. But all it takes to convert it to a geometric one is the realisation that for any two complex numbers z_1 and z_2 , $|z_1 - z_2|$ is simply the distance between the points, say P and Q which represent them in the Argand diagram. (See the accompanying figure.)



As $|z_1| = \sqrt{9 + 16} = 5$, P lies on a circle of radius 5 centred at the origin O . Q moves on a concentric circle of radius 2. So, Q will be closest to P when it is at A and farthest when it is at B , where A and B are the closer and the farther ends of the diameter of the inner circle passing through P . So the maximum distance is $5 + 2 = 7$ while the minimum distance is $5 - 2 = 3$.

This problem is not as trivial as the last one. But once the key idea strikes, the computations involved are minimal. If this idea does not strike, the maximum and the minimum distance can be determined using calculus. Since $|z_2| = 2$, we can take Q as $(2 \cos \theta, 2 \sin \theta)$ where $\theta \in [0, 2\pi]$. The distance between P and Q is $f(\theta)$, where

$$f(\theta) = \sqrt{(2 \cos \theta - 3)^2 + (2 \sin \theta - 4)^2} \quad (1)$$

As $f(\theta)$ is positive, we maximise and minimise $g(\theta) = (f(\theta))^2$ which is a little easier.

$$g(\theta) = 29 - 12 \cos \theta - 16 \sin \theta \quad (2)$$

We can now take $g'(\theta)$. But as in the last question, derivatives are not mandatory here. The expression $12 \cos \theta + 16 \sin \theta$ equals $20(\frac{3}{5} \cos \theta + \frac{4}{5} \sin \theta) = 20(\cos \alpha \cos \theta + \sin \alpha \sin \theta)$ where $\alpha = \tan^{-1}(\frac{4}{3})$. As a result,

$$g(\theta) = 29 - 20 \cos(\theta - \alpha) \quad (3)$$

whose maximum value is $29+20 = 49$ and minimum value is $29-20 = 9$. Correspondingly, the maximum and the minimum of $f(\theta)$ are 7 and 3 respectively. (We are not interested in the points at which these extrema occur. But they are $\theta = \alpha + \pi$ and $\theta = \alpha$ respectively. The corresponding points on the inner circle are B and A , i.e. the two ends of the diameter passing through the point P . This tallies with what we earlier got simply by looking at the figure.)

Yet another method is to apply the Cauchy-Schwarz inequality because of which

$$|12 \cos \theta + 16 \sin \theta| \leq \sqrt{144 + 256} \sqrt{\cos^2 \theta + \sin^2 \theta} = 20 \quad (4)$$

Hence

$$-20 \leq 12 \cos \theta + 16 \sin \theta \leq 20 \quad (5)$$

Substituting this into (2) we get the maximum and the minimum of $g(\theta)$ as 49 and 9 respectively.

So, there are several ways to avoid calculus. To make calculus mandatory, the inner circle could have been replaced by an ellipse. In that case the problem would be in the same class as the JEE 1987 question solved in Comment No. 14 of Chapter 13. But then that would be a full length question as it was in 1987. Now that the JEE is totally MCQ type and the paper setters are forced to ask only questions carrying a prescribed small credit, such problems are often not asked. When they are, the work demanded is unreasonable. With the policy of two papers, one MCQ and the other conventional, it is possible for the ISI paper setters to ask in Paper 1 questions demanding only simple calculations and save the really long ones for Paper 2.

Q.8 Consider two distinct arithmetic progressions (AP) each of which has a positive first term and a positive common difference. Let S_n and T_n be the sums of the first terms of these AP. Then $\lim_{n \rightarrow \infty} \frac{S_n}{T_n}$ equal

- (A) ∞ or 0 depending on which AP has larger first term.
- (B) ∞ or 0 depending upon which AP has larger common difference.
- (C) the ratio of the first terms of the AP.
- (D) the ratio of the common differences of the AP.

Answer and Comments: (D). This is yet another problem where the correct answer can be guessed at by working out one or two particular examples. But the formulas for the sums of the first n terms of an AP are so standard that we might as well do the general case. So let a, b be the first terms and d, e the common differences of the two progressions. Then,

$$S_n = na + \frac{n(n-1)}{2}d \quad (1)$$

$$\text{and } T_n = nb + \frac{n(n-1)}{2}e \quad (2)$$

So,

$$\begin{aligned} \frac{S_n}{T_n} &= \frac{n(n-1)d + 2an}{n(n-1)e + 2bn} \\ &= \frac{dn^2 + (2a-d)n}{en^2 + (2b-e)n} \\ &= \frac{dn + (2a-d)}{en + (2b-e)} \\ &= \frac{d + \frac{2a-d}{n}}{e + \frac{2b-e}{n}} \end{aligned} \quad (3)$$

As $n \rightarrow \infty$, the numerator and the denominator tend to d and e respectively. Hence the ratio tends to $\frac{d}{e}$, which is (D).

The essential idea in the problem is merely that the sum of the first n terms of an AP with common difference d is a quadratic polynomial

in n whose leading coefficient is $\frac{d}{2}$. Once this is realised the problem is a special case of Exercise (6.41) about the limit of a ratio of two polynomials in n . As the result of Exercise (6.41) is fairly well known, good candidates will have to spend very little time to get the answer. A more challenging problem would have to been to ask, for a given positive integer r , the unique integer k for which $\frac{1^r + 2^r + \dots + n^r}{n^{k+1}}$ tends to a finite non-zero limit and also to find this limit. (Exercise (17.11).) But then again, that would be suitable for Paper 2. So, as long as the entire selection is not going to be based solely on Paper 1, it is all right to include in it such questions which demand very little work once you hit the key idea. (The last two questions were also of this category.)

Q.9 Let $f(x) = \max\{\cos x, x^2\}$, $0 < x < \frac{\pi}{2}$. If x_0 is the solution of the equation $\cos x = x^2$ in $(0, \frac{\pi}{2})$, then

- (A) f is continuous only at x_0 .
- (B) f is not continuous at x_0 .
- (C) f is continuous everywhere and differentiable only at x_0 .
- (D) f is differentiable everywhere except at x_0 .

Answer and Comments: (D). The first two options are rendered false by the observation that the maximum of any two continuous functions, say, $f_1(x)$ and $f_2(x)$, defined on a common domain is continuous at every point of that domain. This follows from the simple formula

$$\max\{f_1(x), f_2(x)\} = \frac{f_1(x) + f_2(x) + |f_1(x) - f_2(x)|}{2} \quad (1)$$

and the continuity of the absolute value function $|x|$ of x . (Note that $|x|$ is itself the maximum of x and $-x$ both of which are continuous functions of x . But applying the observation above to prove this would be a case of a **vicious cycle**, i.e. an argument where you set out to prove something and in your proof you use something whose proof requires what you set out to prove in the first place! So, continuity of $|x|$ has to be established independently. But once this is done, the formula above shows that the maximum of any two continuous functions

is continuous. (An analogous result is true for the minimum of two continuous functions.)

As for differentiability too, we can take a cue from the particular example of the function $|x|$. Here both x and $-x$ are differentiable everywhere (with derivatives 1 and -1 at all points respectively). These two functions are equal only at 0. At this point, $|x|$ equals x on the right of it and so the right handed derivative is 1. But $|x|$ equals $-x$ on the left of 0 and so the left handed derivative of $|x|$ at 0 is -1 . As the two sided derivatives are different, $|x|$ fails to be differentiable at 0. But it is differentiable at every $a \neq 0$ because in a sufficiently small neighbourhood of a , $|x|$ equals either x or $-x$ throughout, depending upon whether $a > 0$ or $a < 0$.

These simple ideas are all we need to check differentiability of the function $f(x) = \max\{\cos x, x^2\}$. Let us first make sure that the equation $\cos x = x^2$ has a unique solution in the interval $(0, \frac{\pi}{2})$. For this we consider the difference function $g(x) = \cos x - x^2$. Then $g(0) = 1 > 0$ while $g(\frac{\pi}{2}) = -\frac{\pi^2}{4} < 0$. So by the Intermediate Value Property, $f(x)$ vanishes at least once in the interval $(0, \frac{\pi}{2})$. Further $g'(x) = -\sin x - 2x < 0$ for all $x \in (0, \frac{\pi}{2})$. Hence $g(x)$ is strictly decreasing in this interval. So x_0 is the only solution.

Now to check differentiability of $f(x)$ at x_0 we find the derivatives of the functions $\cos x$ and x^2 at x_0 and see if they are equal. That is, we check whether $-\sin x_0 = 2x_0$. This is obviously not the case as the L.H.S. is negative while the R.H.S. is positive. So, $f(x)$ is not differentiable at x_0 . However, for every other point x_1 of $(0, \frac{\pi}{2})$, there is some neighbourhood of x_1 in which either $\cos x > x^2$ throughout or $\cos x < x^2$ throughout. As a result, in this neighbourhood, $f(x)$ equals either $\cos x$ or x^2 throughout. As both these functions are differentiable at x_1 , so is $f(x)$. Hence (D) is correct.

The essential idea in the question is that when a function $f(x)$ is defined as the maximum of two differentiable functions, say $f_1(x)$ and $f_2(x)$, then at a point x_0 where $f_1(x_0) = f_2(x_0)$, in general $f_1(x) > f_2(x)$ on one side of x_0 and $f_1(x) < f_2(x)$ on the other. Regardless of which possibility holds where, $f(x)$ will fail to be differentiable at x_0 if $f'_1(x_0) \neq f'_2(x_0)$.

Since the question does not ask for a justification, those who

have this essential idea can score quickly because it is *not* necessary to identify the point x_0 . In fact, the graphs of the functions $y = \cos x$ and $y = x^2$ are so standard that even a qualitative sketch will show that at the point of intersection, they have different slopes, without actually identifying the point of intersection. Nor is there any need to check if such a point exists and is unique. In a conventional examination, several justifications could be asked including proving the existence and uniqueness of x_0 .

Q.10 The set of all real numbers in $(-2, 2)$ satisfying

$$2^{|x|} - |2^{x-1} - 1| = 2^{x-1} + 1$$

is

- (A) $\{-1, 1\}$. (B) $\{-1\} \cup [1, 2)$.
 (C) $(-2, -1] \cup [1, 2)$. (D) $(-2, -1] \cup \{1\}$.

Answer and Comments: (B). By including the number -1 in all options, the paper setters have made it unnecessary to check if $x = -1$ is really a solution. (If it is not, then none of the options is correct.) Still, for the sake of honesty, we see that for $x = -1$, the L.H.S. is $2 - |\frac{1}{4} - 1| = 2 - \frac{3}{4} = \frac{5}{4}$ while the R.H.S. is $\frac{1}{4} + 1 = \frac{5}{4}$. So -1 indeed satisfies the given equation.

For other values of x , $|2^{x-1} - 1| = 2^{x-1} - 1$ for $x \geq 1$ and the equation reduces to

$$2^x = 2^{x-1} - 1 + 2^{x-1} + 1 = 2 \times 2^{x-1} \quad (1)$$

which is true for all $x \geq 1$. Hence all points of $[1, 2)$ are in the solution set.

Now consider what happens for $x < 1$. Then $2^{x-1} < 1$ and so $|2^{x-1} - 1| = 1 - 2^{x-1}$ and the given equation reduces to

$$2^{|x|} = 2^{x-1} + 1 + 1 - 2^{x-1} = 2 \quad (2)$$

which will hold if and only if $|x| = 1$. We are already assuming $x < 1$. So the only permissible value is -1 . Hence the solution set is $\{-1\} \cup [1, 2)$. Therefore (B) holds.

The problem is a straight replica of a discarded 1997 JEE question except that the original question asked for all real solutions and not just those in $(-2, 2)$. (The reason for discarding was that the question paper was leaked before the examination.) It is given as Exercise (6.25)(e). But in the solutions given, the point -1 is missed (but corrected in the Errata).

- Q.11 Let $S(k)$ denote the set of all one-to-one and onto functions from $\{1, 2, \dots, k\}$ to itself. Let p, q be positive integers. Let $S(p, q)$ be the set of all τ in $S(p + q)$ such that $\tau(1) < \tau(2) < \dots < \tau(p)$ and $\tau(p + 1) < \tau(p + 2) < \dots < \tau(p + q)$. The number of elements in the set $S(13, 29)$ is

$$(A) 377. \quad (B) (42)!. \quad (C) \binom{42}{13}. \quad (D) \frac{42!}{29!}$$

Answer and Comments: (C). There is, of course, nothing special about either 13 or 29. (Both are prime numbers. But that has little bearing on the problem.) So we might as well calculate $S(p, q)$ for arbitrary positive integers p and q . Every bijection, say τ , from $\{1, 2, \dots, p + q\}$ to itself corresponds to a permutation $\tau(1), \tau(2), \dots, \tau(p + q)$ of the integers $1, 2, \dots, p + q$. In every such arrangement, p positions are filled by the elements $\tau(1), \tau(2), \dots, \tau(p)$ and the remaining q positions are filled by the elements $\tau(p + 1), \tau(p + 2), \dots, \tau(p + q)$. We want only those permutations in which both these sequences are in an ascending order. As a result, the moment we choose which p of the $p + q$ places be occupied by $\tau(1), \tau(2), \dots, \tau(p)$, we can place them at these positions in only one way. Nor do we have any freedom to place the remaining q elements $\tau(p + 1), \tau(p + 2), \dots, \tau(p + q)$ since they have to be put into the remaining q places in an ascending order and there is only one way to do it. So the only choice we have is to decide which p of the $p + q$ places be filled in by $\tau(1), \tau(2), \dots, \tau(p)$. There are $\binom{p + q}{p}$ ways to do this. So (C) is correct.

In case you find this too abstract, consider a class with p boys and q girls all of different heights. We want to seat them in a row so that all the boys are seated in an ascending order of height and so is the case with girls. Then the arrangement is completely determined by which

seats are to be occupied by the boys.

Yet another question where there is very little computation to be done once the key idea strikes.

Q.12 Suppose that both the roots of the equation $x^2 + ax + 2016 = 0$ are positive even integers. The number of possible values of a is

- (A) 6. (B) 12. (C) 18. (D) 24.

Answer and Comments: (B). Call the roots as α and β . Then we have $\alpha\beta = 2016$ and $a = -(\alpha + \beta)$. So the question boils down to how many factorisations of 2016 there are in which both the factors are even positive integers. Letting $\alpha = 2u$ and $\beta = 2v$, this amounts to counting the number of possible factorisations of 504 as products of two positive integers. The prime power factorisation of 504 is $2^3 3^2 7^1$. So every factor of 504 is of the form $2^x 3^y 7^z$ where x, y, z are integers with $0 \leq x \leq 3, 0 \leq y \leq 2$ and $0 \leq z \leq 1$. So, in all 504 has $4 \times 3 \times 2 = 24$ factorisations. But the factorisations $u \times v$ and $v \times u$ give the same value of a (viz. $-(2u + 2v)$). Moreover there are no factorisations in which $u = v$ since 504 is not a perfect square. So the number of possible values of a is not 24 but half of it, i.e. 12.

The problem is more about number theory (specifically, unique factorisation of positive integers) than on quadratic equations. It could have been a little more testing by giving some even perfect square (e.g. 3600) instead of 2016 (which serves little purpose than to conform to a recent fashion in annual examinations to set one problem which gives the year of the examination). It is also important to note that if we ignore the order of the factors, then no two distinct factorisations of 504 will give the same value of a . This is not as automatic as it appears. And, indeed, a similar statement is false if, instead of a factorisation into two factors we consider a factorisation into three factors. For example, 36 can be factorised either as $9 \times 2 \times 2$ or as $6 \times 6 \times 1$, in both of which the factors add to the same sum, viz. 13. That this can't happen when there are only two factors, needs a proof. Not that the proof is hard. Indeed, suppose $504 = u_1 v_1 = u_2 v_2$ and $u_1 + v_1 = u_2 + v_2$. Then u_2 and v_2 are the roots of the same quadratic which has got u_1 and v_1 as its roots. But then either $u_2 = u_1$ (whence $v_1 = v_2$) or

$u_2 = v_1$ (whence $u_1 = v_2$). So, except for the order of the two factors, the factorisations are the same.

Once again, it is only in a conventional type examination that it can be tested whether a candidate has addressed these fine points.

Q.13 Let $b \neq 0$ be a fixed real number. Consider the family of parabolas given by the equations

$$y^2 = 4ax + b, \quad \text{where } a \in \mathbb{R}.$$

The locus of the points on the parabolas at which the tangents to the parabolas make 45° angle with the x -axis is

- | | |
|----------------------|-------------------------------|
| (A) a straight line. | (B) a pair of straight lines. |
| (C) a parabola. | (D) a hyperbola. |

Answer and Comments: (D). By the angle the tangent makes with the x -axis, we take it to mean the angle with the positive x -axis. Then the condition about the tangent making 45° angle with the x -axis simply means that its slope is 1. The given equation is a one parameter family of parabolas, the parameter being a (and not b , which is a fixed constant). On each member of this family there will be one point at which the tangent has slope 1. We have to find the locus of this point as a varies.

As usual, we let (x_0, y_0) be a typical point on the locus. Then we first have

$$y_0^2 = 4ax_0 + b \tag{1}$$

for some $a \in \mathbb{R}$. The slope of the tangent to the parabola $y^2 = 4ax + b$ at the point (x_0, y_0) is the value of $\frac{dy}{dx}$ at (x_0, y_0) . From the equation of the parabola, we get

$$2y \frac{dy}{dx} = 4a \tag{2}$$

We are given that at (x_0, y_0) , $\frac{dy}{dx} = 1$. So we get

$$y_0 = 2a \tag{3}$$

(An alternate derivation is to write down the equation of the tangent to the parabola $y^2 = 4ax + b$ at the point (x_0, y_0) as $yy_0 = 2a(x + x_0)$ and equate its slope with 1.)

From (3) it is tempting to think that the locus is the line $y = 2a$. But that is wrong, because a is not a constant but a parameter. To obtain the locus of (x_0, y_0) , we eliminate the parameter a from (1) and (3). This gives

$$y_0^2 = 2x_0y_0 + b \quad (4)$$

Replacing x_0, y_0 by x, y respectively we get the locus as

$$y^2 = 2xy + b \quad (5)$$

or, in a standard form,

$$Ax^2 + 2Hxy + By^2 = 1 \quad (6)$$

where $A = 0, H = -\frac{1}{b}$ and $B = \frac{1}{b}$. This is a hyperbola because $H^2 - AB = \frac{1}{b^2} - 0 > 0$. So (D) is true. (In going from (5) to (6), we have used that $b \neq 0$. If $b = 0$, then (5) represents a pair of straight lines, $y = 0$ and $y = 2x$. But no point on or below the x -axis can be on the locus since the parabolas cut the x -axis only at the origin, at which the tangent is vertical, while the tangents at points below the x -axis have negative slopes. So, in this case the locus is the portion of the line $y = 2x$ in the first quadrant. Often such a set is called a **ray** or **half ray**. It is precisely the set of the upper extremities of the latus recti of the parabolas.)

If we allow the tangent to make an angle 45° with the negative x -axis, we would have to include the points where the tangent has slope -1 too. The work will be the same but the locus will come out to be

$$y^2 = \pm 2xy + b \quad (7)$$

which is a union of two hyperbolas. But this is not one of the given options. It would have been far better if the paper setters had simply said that the tangent has slope 1. This would avoid ambiguity without giving any unwarranted help to a candidate.

Q.14 Consider the curve represented by the equation

$$ax^2 + 2bxy + cy^2 + d = 0$$

in the plane, where $a > 0, c > 0$ and $ac > b^2$. Suppose that the normals to the curve drawn at 5 distinct points on the curve all pass through the origin. Then

- | | |
|------------------------------|------------------------------|
| (A) $a = c$ and $b > 0$. | (B) $a \neq c$ and $b = 0$. |
| (C) $a \neq c$ and $b < 0$. | (D) None of the above. |

Answer and Comments: (D). Points on a curve such that the normals to the curve at them are concurrent are called **conormal points**. The question is based on the knowledge of how many conormal points a conic can have. In the extreme case of a circle all points are conormal, since all the normals pass through the centre of the circle. However, for other conics there can be at most four conormal points. For example, in the case of an ellipse the extremities of the two axes are conormal points. For a parabola, from any point not on it, in general three normals can be drawn. The points where these three normals meet the parabola are conormal points.

The proof that except for a circle no conic can have more than four conormal points meeting at a point (x_0, y_0) consists of taking a typical line $y - y_0 = m(x - x_0)$ through this point and writing down the condition for it to be a normal to the conic. This condition will be in the form of an equation in m . In the case of a circle, this equation degenerates into $0 = 0$ when (x_0, y_0) is the centre of the circle. But in other cases it turns out to be a polynomial equation of degree at most four. A candidate can hardly be expected to do all this work on the spot. So this is a question where previous knowledge of a relatively less common fact is crucial. Once you know this crucial fact, all that remains to do is to see if the conic is a circle centred at the origin. The general equation of such a circle is $ax^2 + ay^2 + d = 0$ where $ad < 0$ (i.e. a and d are of opposite signs). None of the options (A), (B), (C) meet this and so (D) is correct.

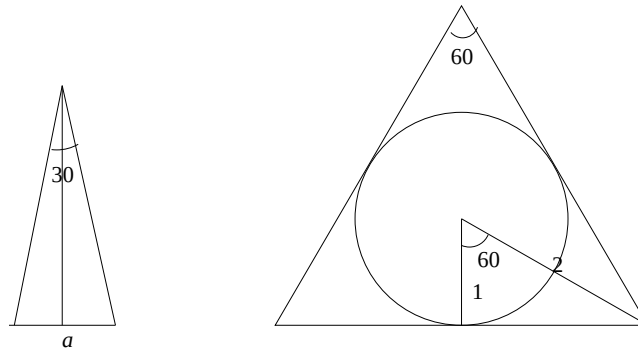
A clever candidate who cannot recall the fact about conormal points can still guess the answer because of the unusual option ‘None of the

above'. There was a time when this option was used as a safeguard in case none of the other options is correct because of some slip in the problem. That practice is now discouraged as it sometimes may be misused. So, today, if this option appears, then it is intentional and then the chances are that it is indeed correct!

- Q.15 Let P be a 12-sided regular polygon and T be an equilateral triangle with its incircle having radius 1. If the area of P is the same as the area of T , then the length of the side of P is

- (A) $\sqrt{\sqrt{3} \cot 15^\circ}$ (B) $\sqrt{\sqrt{3} \tan 15^\circ}$
 (C) $\sqrt{3\sqrt{2} \tan 15^\circ}$ (D) $\sqrt{3\sqrt{2} \cot 15^\circ}$

Answer and Comments: (B). A straightforward problem about the calculation of the areas of two figures, one a regular 12-gon and the other an equilateral triangle. Let a be the side of the 12-gon. Divide the 12-gon into 12 equal isosceles triangles by joining its centre to the vertices. One such triangle is shown in the figure below along with an equilateral triangle with inradius 1.



Clearly, the area of the 12-gon, say A equals

$$A = 12 \times \frac{a}{2} \times \frac{a}{2} \cot 15^\circ = 3a^2 \cot 15^\circ \quad (1)$$

On the other hand, if we divide the equilateral triangle into six triangles of which one is shown in the figure, then its area, say B , is

$$B = 6 \times \frac{1}{2} \times 1 \times 2 \times \sin 60^\circ = 3\sqrt{3} \quad (2)$$

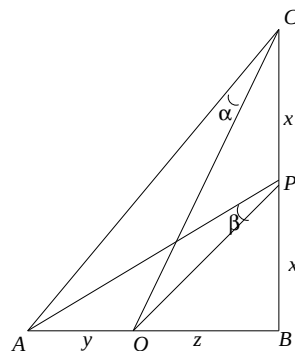
Equating A with B gives $a^2 = \sqrt{3} \tan 15^\circ$. So (B) is correct.

An extremely straightforward problem. But in such problems a candidate is likely to commit a numerical slip (e.g. forgetting the factor $\frac{1}{2}$ in calculating the area of a triangle). To some extent, the MCQ format gives an alert against such errors.

- Q.16 Let ABC be a right-angled triangle with $\angle ABC = 90^\circ$. Let P be the midpoint of BC and Q be a point on AB . Suppose that the length of BC is $2x$, $\angle ACQ = \alpha$, and $\angle APQ = \beta$. Then the length of AQ is

- (A) $\frac{3x}{2 \cot \alpha - \cot \beta}$. (B) $\frac{2x}{3 \cot \alpha - 2 \cot \beta}$.
 (C) $\frac{3x}{\cot \alpha - 2 \cot \beta}$. (D) $\frac{2x}{2 \cot \alpha - 3 \cot \beta}$.

Answer and Comments: (A). Another trigonometry problem. But far from straightforward. Problems of this type were common when ‘Heights and Distances’ was a part of the JEE syllabus (and hence of other similar examinations). In fact the present problem could have been cast in the form, ‘An observer walks horizontally towards the base of a vertical lamp post of height $2x$. When he has covered a distance y , he notices that the path travelled subtends an angle α at the top of the lamp post and an angle β at the midpoint of the post. Shew that $y = \dots$ ’



‘Heights and distances’ have been removed from the JEE syllabus possibly because candidates poor in languages had difficulties understanding the problems, even though they were mathematically capable. But this was like throwing the baby with the bath water, because an important ability to convert a real life data into a mathematical problem could no longer be tested. So what we have here is a ready to cook version in which you don’t worry what BC and AQ stand for.

But even though the old world charm is gone, the problem is still an interesting one. The important thing to observe is the similarity of the data about the angles α and β . We are interested in the length of AQ . Let us call it y . We introduce an auxiliary variable z equal to the length QB . Then

$$\angle ACB = \tan^{-1}\left(\frac{y+z}{2x}\right) \text{ and } \angle QCB = \tan^{-1}\left(\frac{z}{2x}\right) \quad (1)$$

Now, $\alpha = \angle ACQ = \angle ACB - \angle QCB$. So the two equations above imply

$$\begin{aligned} \tan \alpha &= \tan(\angle ACB - \angle QCB) \\ &= \frac{\frac{y+z}{2x} - \frac{z}{2x}}{1 + \frac{(y+z)z}{4x^2}} = \frac{2xy}{4x^2 + (y+z)z} \end{aligned} \quad (2)$$

In an entirely analogous manner, $\beta = \angle APB - \angle QPB$ gives

$$\tan \beta = \frac{\frac{y+z}{x} - \frac{z}{x}}{1 + \frac{(y+z)z}{x^2}} = \frac{xy}{x^2 + (y+z)z} \quad (3)$$

The desired formula for y can be obtained by eliminating z (which was an auxiliary variable of our choice) from (2) and (3). This is very easy. From (2),

$$(y+z)z = 2xy \cot \alpha - 4x^2 \quad (4)$$

while (3) gives

$$(y+z)z = xy \cot \beta - x^2 \quad (5)$$

Equating and cancelling x we get

$$2y \cot \alpha - 4x = y \cot \beta - x \quad (6)$$

which implies

$$y = \frac{3x}{2 \cot \alpha - \cot \beta} \quad (7)$$

Since $y = AQ$, (A) is correct.

This is an excellent problem. Although the knowledge of trigonometry needed is elementary (the formula for the tangent of the difference of two angles), art lies in making good use of it after introducing an auxiliary variable z . Such variables which are not a part of the data are like catalysts in chemical reactions.

Q.17 Let $[x]$ denote the greatest integer less than or equal to x . The value of the integral

$$\int_1^n [x]^{x-[x]} dx$$

is equal to

- (A) $1 + \frac{2^3}{\log_e 2} - \frac{2^2}{\log_e 2} + \frac{3^4}{\log_e 3} - \frac{3^3}{\log_e 3} + \dots + \frac{(n-1)^n}{\log_e(n-1)} - \frac{(n-1)^{n-1}}{\log_e(n-1)}.$
- (B) $1 + \frac{1}{\log_e 2} + \frac{2}{\log_e 3} + \dots + \frac{(n-2)}{\log_e(n-1)}.$
- (C) $\frac{1}{2} + \frac{2^2}{3} + \dots + \frac{n^{n+1}}{n+1}.$
- (D) $\frac{2^3-1}{3} + \frac{3^4-2^3}{4} + \dots + \frac{n^{n+1}-(n-1)^n}{n+1}.$

Answer and Comments: (B). The expression in each option is a sum of $n-1$ terms. (In (A), it is a sum of $2n-3$ terms. But if we add -0 as a dummy second term then after pairing, it is a sum of $n-1$ terms. In (C), there are n terms, probably because of a misprint.) The given integral, say I , is also the sum of $n-1$ integrals $I_1 + I_2 + \dots + I_{n-1}$ where for every $k = 1, 2, \dots, n-1$ we let

$$I_k = \int_k^{k+1} [x]^{x-[x]} dx \quad (1)$$

So, we evaluate I_k and check in which option it matches with the k -th term. On the interval $[k, k+1]$, $[x] \equiv k$. (At the end point $k+1$, $[x] = k+1$. But changing the value of the integrand at one, and indeed, at any finite number of points does not change the integral.) The exponent $x-[x]$ is the fractional part of x and varies from 0 to 1 as $x \in [k, k+1]$. So, if we substitute $t = x - k$ and write $\ln k$ instead of $\log_e k$ which is a standard practice now, then

$$\begin{aligned} I_k &= \int_0^1 k^t dt \\ &= \int_0^1 e^{t \ln k} dt \\ &= \frac{1}{\ln k} k^t \Big|_0^1 = \frac{k-1}{\ln k} \end{aligned} \quad (2)$$

which matches with the k -th term of the series in (B). (For $k = 1$, there is a slight degeneracy because $\ln 1 = 0$. But in this case, the integrand of I_1 is a constant 1 and so the integral is 1.)

The statement of the problem ought to have included that n is a positive integer, although that is clear from the context. The trouble with many MCQ's involving an integer parameter n is that the correct option can often be identified by working out only some simple cases such as $n = 1$ or $n = 2$. Such questions are, therefore, not suitable as MCQ's.

Q.18 Let $\alpha > 0, \beta \geq 0$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous at 0 with $f(0) = \beta$. If $g(x) = |x|^\alpha f(x)$ is differentiable at 0, then

- | | |
|---------------------------------------|--|
| (A) $\alpha = 1$ and $\beta = 1$. | (B) $0 < \alpha < 1$ and $\beta = 0$. |
| (C) $\alpha \geq 1$ and $\beta = 0$. | (D) $\alpha > 0$ and $\beta > 0$. |

Answer and Comments: (None). The function $|x|$ is, by far, the most standard example of a function which is continuous but not differentiable at 0. In the comments on Q.9 we elaborated why it fails to be differentiable at 0. However, if $\alpha > 1$, then $g(x) = |x|^\alpha$ is differentiable at 0. This follows if we write $\alpha = 1 + u$ where $u > 0$. Then for $x \neq 0$, $\frac{g(x) - g(0)}{x} = \frac{|x|}{x} \times |x|^u$. The first factor tends to 1 or -1 depending upon whether $x \rightarrow 0$ from the right or the left. But it is bounded and so because of the second factor $|x|^u$, $\frac{g(x)}{x}$ tends to 0 as x tends to 0 from either side.

Put differently, the non-differentiability of $|x|$ at 0 gets 'cured' or 'masked' when multiplied by a factor which tends to 0 as $x \rightarrow 0$. This other factor need not be some power of $|x|$. In the present problem, $f(x)$ is given to be continuous at 0 and $f(0) = \beta$. So, $f(x) \rightarrow \beta$ as $x \rightarrow 0$. Therefore $f(x)$ will work as a curing factor if $\beta = 0$. But if $\beta \neq 0$ and $\alpha = 1$, then $|x|f(x)$ will be non-differentiable at 0 since the right handed derivative will be β while the left handed derivative will be $-\beta$. So for differentiability of $|x|^\alpha f(x)$ at 0, it would suffice if either $\alpha > 1$ or ($\alpha = 1$ and $\beta = 0$.) Logically this is not the same as (C) which says that both $\alpha \geq 1$ and $\beta = 0$ should hold. But among the

given options, it is the closest. (In fact, even when $0 < \alpha < 1$, $|x|^\alpha f(x)$ will be differentiable at 0 if the factor $f(x)$ tends to 0 more rapidly than $|x|^{1-\alpha}$, e.g. $|x|^{1/2} \sin^2 x$ is differentiable at 0. What matters is not just that $f(x)$ tends 0 as $x \rightarrow 0$ but that it does so sufficiently rapidly. This seems to have been missed.)

Q.19 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable and strictly decreasing function such that $f(0) = 1$ and $f(1) = 0$. For $x \in \mathbb{R}$, let

$$F(x) = \int_0^x (t - 2)f(t)dt.$$

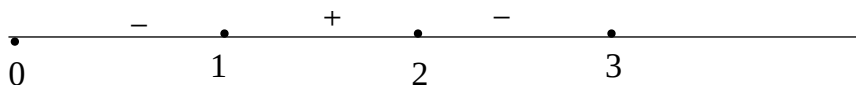
Then

- (A) F is strictly increasing in $[0, 3]$.
- (B) F has a unique minimum in $(0, 3)$ but has no maximum in $(0, 3)$.
- (C) F has a unique maximum in $(0, 3)$ but has no minimum in $(0, 3)$.
- (D) F has a unique maximum and a unique minimum in $(0, 3)$.

Answer and Comments: (D). All the options can be tested using $F'(x)$ provided it exists. This is guaranteed by the fundamental theorem of calculus. So we have

$$F'(x) = (x - 2)f(x) \tag{1}$$

for all $x \in \mathbb{R}$. The first factor is negative for $x < 2$ and positive for $x > 2$. The second factor is positive for $x < 1$ and negative for $x > 1$ because it is given that $f(1) = 0$ and f is strictly decreasing. So the signs of $F'(x)$ for $x \in (0, 3)$ will be as shown below.



So, F has a local minimum at 1 and a local maximum at 2. Whether these are the global extrema for $f(x)$ on $(0, 3)$ will depend on $f(x)$ and the data is insufficient to conclude anything. If we interpret local extrema as extrema, then (D) is correct. But this is dangerous. (JEE 2008 invited criticism for the same lapse.) Many calculus books and

teachers painstakingly point out the difference between a local maximum and a maximum. A candidate so trained will get confused.

As a possible attempt to salvage the situation, it is clear that $F(0) = 0$. If it can be shown from the data that $F(2) \geq 0$, then F would have a global maximum at 2 on the interval $[0, 3]$ and hence on the smaller interval $(0, 3)$ too. Apparently this is what the paper setters had in mind as otherwise there is little point in defining $F(x)$ by an integral. If the idea was to test the knowledge of the fundamental theorem of calculus, mere continuity of f would suffice. It is not clear why f is given to be differentiable. The piece of data $f(0) = 1$ also seems irrelevant. Using it, one can indeed show that $F(1) = \int_0^1 (t-2)f(t)dt$ lies between -2 and -1 since the integrand lies between these two numbers on $[0, 1]$. We can then write $F(2)$ as $F(1) + \int_1^2 (t-2)f(t)dt$. If the second term can be shown to be at least 1, then from $F(1) \geq -1$ we would indeed get that $F(2) \geq -1 + 1 = 0 = F(0)$. Apparently this is what the paper setters had in mind. But in absence of any additional information about the values of $f(t)$ for $1 \leq t \leq 2$, little can be concluded by way of a lower bound for $\int_1^2 (t-2)f(t)dt$. Even though both the factors are negative, the second one could be only marginally negative.

Summing up, like the last problem, this one is also likely to confuse the scrupulous candidates.

Q.20 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a non-zero function such that $\lim_{x \rightarrow \infty} \frac{f(xy)}{x^3}$ exists for all $y > 0$. Let $g(y) = \lim_{x \rightarrow \infty} \frac{f(xy)}{x^3}$. If $g(1) = 1$, then for all $y > 0$

(A) $g(y) = 1$. (B) $g(y) = y$. (C) $g(y) = y^2$. (D) $g(y) = y^3$.

Answer and Comments: (D). An intuitive argument can be given as follows. We are given that $g(1) = 1$. This means

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x^3} = 1 \tag{1}$$

This means that for large x , $f(x)$ is comparable to x^3 . But in that

case $f(xy)$ will be comparable to $(xy)^3$, i.e. x^3y^3 . So the ratio $\frac{f(xy)}{x^3}$ is comparable to y^3 . So the limit $g(y)$ equals y^3 .

For a rigorous proof, we divide and multiply by $(xy)^3$ to get

$$\frac{f(xy)}{x^3} = \frac{f(xy)}{(xy)^3} \times \frac{x^3y^3}{x^3} \quad (2)$$

For a fixed $y > 0$, $xy \rightarrow \infty$ as $x \rightarrow \infty$. So the first factor tends to 1 by hypothesis. The second factor is y^3 . Hence $g(y) = y^3$ for all $y > 0$.

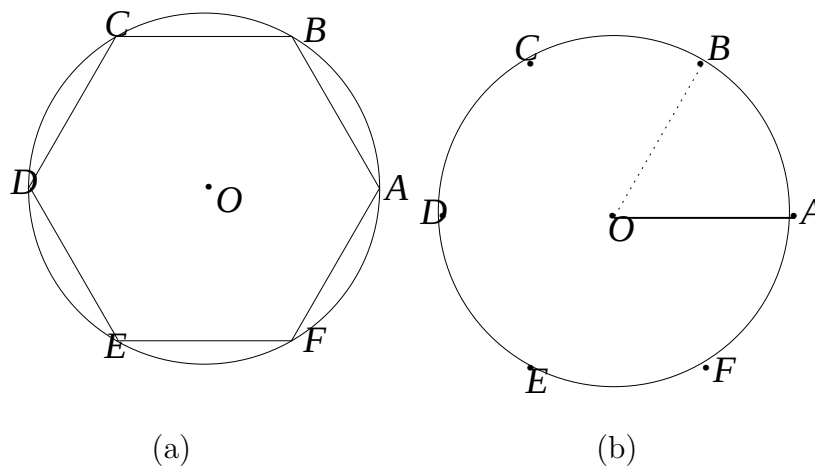
The hypothesis that f is non-zero is redundant since it is given that $f(x)/x^3 \rightarrow 1$ as $x \rightarrow \infty$.

A good problem which can be done by an intuitive argument too. Unfortunately, it can also be done by taking a very special case, viz. $f(x) = x^3$. Then all conditions are satisfied and at least in this case $g(y) = y^3$. As only one option is given to be correct, it must be (D).

- Q.21 Let $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$. Then the maximum number of points in D such that the distance between any pair of points is at least 1 will be

(A) 5. (B) 6. (C) 7. (D) 8.

Answer and Comments: (C). The given set D is a unit disc i.e. a disc of radius 1. We want to put as many points in it as possible subject to the requirement that no two should be closer than 1 unit of length. The condition means that if P is a selected point, then no point of D which lies in the interior of the disc D' of radius 1 centred at P can be chosen. To minimise this prohibited area (viz. $D \cap D'$), P should lie on the unit circle $S = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$, which is the boundary of the disc D . So intuitively, the circle S is the best place to put these points. Going one step further, for the selected points on the unit circle S , we should not waste space by keeping two adjacent ones at a distance more than 1 apart. In other words, we should take points on S so that every adjacent pair is at a distance exactly 1 apart. This is possible if we let the points be the vertices of a regular hexagon, say $ABCDEF$ inscribed in S . (See (a) of the figure below.)



(Note that D also denotes the unit disc. But the context makes it clear what it stands for. Those who are very fussy about double usage of notations may denote the vertices of the hexagon by A, B, C, E, F, G if they prefer, just as the hotel rooms in a Western superstitious hotel are numbered from 1 to 12 and then from 14 onwards!). So the six vertices of the regular hexagon have the desired property. In addition, the centre, say O of the disc can be included. Thus we get 7 points such that the minimum distance between any two of them is 1. So, if we let $X = \{A, B, C, D, E, F, O\}$, we get a subset of D satisfying the given restriction. Hence the answer to the problem is at least 7.

But how do we know that the answer is not more than 7?. Surely, the set X cannot be enlarged to a bigger set with the same property. For, suppose P is a point of D not in X . Then the requirement that its distance from O is at least 1 forces P to lie on the unit circle S . But then P will have to lie somewhere in between two adjacent vertices which will violate the requirement of minimum distance.

But just because X cannot be enlarged to a bigger subset of D having the said property does not, by itself, mean that nobody can construct a subset with 8 or more elements having the given property. To understand this subtle point, suppose we have a 7 storeyed building. Then it may not be possible to build an eighth floor on its top, as the foundations may not be strong enough. But that does not mean that some other building with 8 storeys cannot be built next to it.

So, how do we show that it is impossible to construct a subset, say Y , of S with 8 elements having the given property? We cannot just go

on trying one subset after another and discarding it. This would have been feasible if the set D were finite. Even then it is not a very efficient method. Anyway, here the question does not arise because the set D is infinite and so there are infinitely many subsets Y of it containing 8 elements.

The proof of non-existence of such a set Y is tricky but based on a very simple principle, called the **pigeon hole principle**. As the name indicates, if we have m pigeon holes and none of them can hold more than one letter, then we cannot put $m + 1$ (or more) letters into these m pigeon holes. This is sheer common sense. But sometimes this principle works wonders. Art lies in designing these ‘pigeon holes’ suitably for a given problem. (See Comment No. 16 of Chapter 6 for some illustrations. The present problem is similar to Exercise (6.36).)

We put this principle to use in the present problem by constructing seven ‘pigeon holes’. Each of these will be a subset of the disc D . Our first pigeon hole will be simply the singleton set $\{O\}$. We now divide the complement $D - \{O\}$ (often suggestively called the **punctured unit disc**) into six mutually disjoint subsets as follows. Consider the regular hexagon $ABCDEF$ above inscribed in the boundary circle S . We divide D into six sectors by joining O to the vertices. These sectors are overlapping in their boundaries. But if we carefully exclude some boundary points from each sector, we can make them mutually disjoint. Consider the sector OAB , shown in (b) of the figure above. We keep all points on the radius OA except O . This is shown by thickening it. We drop all points on the radius OB , including the end points O and B (shown by the dotted line). We retain all other points of the sector. (The idea of such partial dropping of the boundary points is also inherent when we form semi-open intervals from closed intervals.) We convert the remaining five sectors analogously, in a cyclic manner. Thus from the sector OBC we keep the radius OB (except for O) and drop the radius OC entirely. The sixth sector OFA will not contain the radius OA which was included in the first sector. Note that because of the exclusion of some of the boundary points, none of the sectors can contain two points at a distance 1 or more.

These six sectors along with $\{O\}$ are our seven pigeon holes. The set Y can have at most one element from each. So it cannot have more than 7 points. So the answer 7 is correct.

The problem is of an unusual nature and an excellent example of how an intuitive thinking can lead to the right answer. Unfortunately, because of the MCQ format, it cannot be decided whether a candidate who answers it correctly can also give the reasoning for the latter half which is more challenging than the first. This question would be ideal as a full length question.

This problem looks more like a puzzle. But such problems also have real life analogies. For example, a chain of stores wants to issue licenses to retailers so that no two of them are closer to each other than a prescribed distance. What is the maximum number of licenses it can issue for a given locality?

Q.22 The number of 3-digit numbers abc such that we can construct an isosceles triangle with sides a, b and c is

- (A) 153. (B) 163. (C) 165. (D) 183.

Answer and Comments: (C). Let us first count how many unordered triples of digits there are in which at least two entries are the same and the three form the sides of a triangle. Call the equal entries x and the third one y . (It may happen that y also equals x in which case we get an equilateral triangle. By an isosceles triangle, some people exclude an equilateral triangle just as some people do not allow rectangles to be squares. But we shall not do so.)

The restriction on y is that $y < 2x$ as otherwise there will not be a triangle. For $x = 1, 2, 3, 4$ and 5 the number of possible values of y is $1, 3, 5, 7$ and 9 respectively. However, for $x > 5$ we can let y vary only from 1 to 9 as it is a single digit. So the total number of unordered triples $\{x, x, y\}$ is $1 + 3 + 5 + 7 + 9 \times 5 = 61$. Out of these, 9 are of the form $\{x, x, x\}$. Each of these gives rise to only one 3-digit number whose digits are the sides of an equilateral triangle. But each of the remaining 52 triples will be of the form $\{x, x, y\}$ where $y \neq x$. Each such triple will give rise to three 3-digit integers, viz. xyx, xyx and yxx . So the number of 3-digit integers we get from these 52 triples is $52 \times 3 = 156$. Adding 9 (which is the number of integers of the form xxx) we get 165 as the answer.

The problem is a simple counting problem. If we exclude equilateral

triangles, the answer would have been 156. As this is not one of the given options, it has to be inferred that the paper setters wanted to allow equilateral triangles too. It is all a matter of convention. But it is always best to spell it out in the statement of the question.

The problem is strikingly similar to a JEE 2015 Mains problem, where it was given that the three figures on three throws of an ordinary die, form the sides of an isoscles triangle and the probability that the triangle have maximum area was asked. The only difference from the present problem is that each a, b, c can take only values from 1 to 6 instead of from 1 to 9.

Q.23 The function

$$f(x) = x^{1/2} - 3x^{1/3} + 2x^{1/4}, \quad x \geq 0$$

- (A) has more than two zeros.
- (B) is always nonnegative.
- (C) is negative for $0 < x < 1$.
- (D) is one-to-one and onto.

Answer and Comments: (B). It is much easier to handle powers with whole exponents than those with fractional ones. In the present problem, this can be arranged by a substitution $x = y^{12}$ where the figure 12 is the l.c.m. of the denominators of the three exponents. Whenever such a substitution is used, one has to ensure that the new function that results behaves the same way as the original function. (To appreciate this point, with a substitution like $x = y^2$, the new function will always be an even function of y even though the original function may not be an even function of x . So such a substitution is not valid in general.) In the present case we have to check that the new function

$$g(y) = y^6 - 3y^4 + 2y^3 \tag{1}$$

behaves the same way as the original function $f(x)$ does. But that is taken care of here because as we are dealing with only non-negative values of x , the substitution $x = y^{12}$ is one-to-one and onto from $[0, \infty)$ to $[0, \infty)$. So g will vanish at some point c if and only if f will vanish

at the point $c^{1/12}$. Moreover, it is a strictly increasing substitution and so the new function will be increasing/decreasing on an interval (a, b) precisely when the original function is so in the corresponding interval $[a^{1/12}, b^{1/12}]$.

It is an easy matter to study the behaviour of $g(y)$ for $y \geq 0$. Clearly $g(y) = 0$ and $g(1) = 0$. To see if g (and hence f) has any more zeros, we factorise $g(y)$ as

$$g(y) = y^3(y^3 - 3y + 2) \quad (2)$$

The first factor vanishes only at 0. Call the second factor as $h(y)$. Then $h(0) = 2$ and $h'(y) = 3(y^2 - 1) < 0$ for $0 < y < 1$. So, h decreases strictly from $h(0) = 2$ to $h(1) = 0$ and increases strictly for $y > 1$. So h has only one zero for $y > 0$. Hence $g(y)$ and so $f(x)$ too has only two zeros. So (A) is false. From what we have done, $h(y) \geq 0$ for all $y \geq 0$. So $f(x)$ is non-negative for all $x \geq 0$. Thus (B) is correct.

After reducing the problem to a simpler one with a valid substitution, the rest of the work boils down to studying properties of a cubic. A far better question testing the latter was Q.4. So this question is repetitious.

There is also a trickier way of proving that $g(y) \geq 0$ from (1). We rewrite $y^6 + 2y^3$ as $y^6 + y^3 + y^3$ and apply the A.M.-G.M. inequality to get that $y^6 + 2y^3 \geq 3(y^6 y^3 y^3)^{1/3} = 3y^4$. So $g(y) \geq 0$ for all $y \geq 0$. No calculus is needed here. In fact, if you hit this trick, you can apply it equally easily to the original function $f(x) = x^{1/2} - 3x^{1/3} + 2x^{1/4}$ without using the substitution. (A similar trick gives $\sec^2 x + 2 \cos x \geq 3$ and hence $\tan x + 2 \sin x \geq 3x$ for $x \in [0, \pi/2)$. See Comment No. 11 of Chapter 13.)

Q.24 Let $X = \{a + \sqrt{-5}b : a, b \in \mathbb{Z}\}$. An element $x \in X$ is called *special* if there exists $y \in X$ such that $xy = 1$. The number of special elements in X is

- (A) 2. (B) 4. (C) 6. (D) 8.

Answer and Comments: (A). An element $x = a + \sqrt{-5}b$ is the complex number $a + \sqrt{5}ib$. Suppose there is some $y = c + i\sqrt{5}d \in X$

with $xy = 1$. Taking absolute values of both the sides

$$|xy|^2 = |x|^2|y|^2 = (a^2 + 5b^2)(c^2 + 5d^2) = 1^2 = 1 \quad (1)$$

As a, b, c, d are integers and $a^2 + 5b^2$ and $c^2 + 5d^2$ are non-negative the only possibility is $a^2 + 5b^2 = 1$. This forces $b = 0$ and $a = \pm 1$.

A person who has studied some algebra will recognise that the question asks for the number of units in the ring $\mathbb{Z} + i\sqrt{5}\mathbb{Z}$. Candidates are not expected to know this. But a clever use of the right trick (multiplicative property of the absolute value of a complex number) makes the solution. The brute force method would be to equate the real and imaginary parts of the product $(a + i\sqrt{5}b)(c + i\sqrt{5}d)$ with 1 and 0 respectively and look for solutions. This is laborious.

Q.25 For a set X , let $P(X)$ denote the set of all subsets of X . Consider the following statements.

(I) $P(A) \cap P(B) = P(A \cap B)$.

(II) $P(A) \cup P(B) = P(A \cup B)$.

(III) $P(A) = P(B) \Rightarrow A = B$.

(IV) $P(\emptyset) = \emptyset$.

Then

(A) All the statements are true.

(B) (I), (II), (III) are true and (IV) is false.

(C) (I), (III) are true and (II), (IV) are false.

(D) (II), (III), (IV) are true and (I) is false.

Answer and Comments: (C). An elementary problem requiring nothing beyond the definition of a subset, and those of the intersection and the union of two subsets. The set $P(X)$ is called the **power set** of X , for the reason that if X has n elements then $P(X)$ has 2^n elements, as for every element of X , there are two possibilities, either including it or not including it in a subset. But this knowledge is not needed in this problem. Here we simply follow the definitions.

Falsity of (IV) follows because the power set of a subset has at least the empty set $\emptyset$ in it. This rules out options (A) and (D). Since both the remaining options claim (I) and (III) are true and only one of them is correct, we need only check whether (II) is true. It is easily seen that it is false. For example, let $A = \{a\}$ and $B = \{b\}$ where $a \neq b$. Then $A \cup B = \{a, b\} \in P(A \cup B)$. But it is neither a subset of A nor of B . So it is not in $P(A) \cup P(B)$. So (C) is true.

Because of some carelessness in paper setting, we could identify the correct option without checking whether (I) and (III) are true. For an honest answer, we must prove them. (I) follows from the fact for any set S , $S \subset A \cap B$ if and only if $S \subset A$ and $S \subset B$. Finally, for (III), observe that for any set X , X is the largest element of X , in the sense of set inclusion. So, if $P(A)$ and $P(B)$ are equal, their largest elements must be equal too. (A more conventional argument is to note that since $A \in P(A)$, $A \in P(B)$, which means $A \subset B$. Similarly, $B \subset A$. So $A = B$.)

A simple problem requiring pure logic and no computational skills like those in Q.16 or Q.17. As logic is the very heart of mathematics, such problems deserve a place once in a while.

Q.26 Let a, b, c be real numbers such that $a + b + c < 0$. Suppose that the equation $ax^2 + bx + c = 0$ has imaginary roots. Then

- | | |
|---------------------------|---------------------------|
| (A) $a < 0$ and $c < 0$. | (B) $a < 0$ and $c > 0$. |
| (C) $a > 0$ and $c < 0$. | (D) $a > 0$ and $c > 0$. |

Answer and Comments: (A). The existence of imaginary roots is equivalent to saying that $b^2 < 4ac$. As a result, ac is positive and so a and c are either both positive or both negative. So the correct option is either (A) or (D). Nothing is given about b other than $b^2 < 4ac$. If we freely choose $b = 0$, then $a + b + c < 0$ reduces to $a + c < 0$ and since both a, c are of the same sign, they must both be negative. So (A) is the correct choice. So, once again the correct option can be identified without doing an honest work. To really rule out (D), assume a, c are both positive and $a + b + c < 0$. Then $-b > a + c$ and as both the sides are positive this means $(-b)^2 = b^2 > (a + c)^2 = a^2 + c^2 + 2ac$. By

the A.M.-G.M. inequality, $a^2 + c^2 \geq 2ac$. So $b^2 \geq 4ac$, contradicting the hypothesis. (The A.M.-G.M. inequality can be bypassed by writing $(a+c)^2$ as $(a-c)^2 + 4ac$ and observing that the first term is non-negative. But this is no gain, because this is how the A.M.-G.M. inequality is proved.)

The problem can also be solved geometrically. Let $f(x) = ax^2 + bx + c$. Since $f(x)$ has no real roots, the graph of $y = f(x)$ lies entirely above or entirely below the x -axis. But $f(1) = a + b + c$ is given to be negative. So the second possibility holds. But then the graph is a downward parabola which forces $a < 0$. As a and c have the same signs, we get that (A) holds.

Q.27 For $\alpha \in (0, \frac{3}{2})$, define $x_n = (n+1)^\alpha - n^\alpha$. Then $\lim_{n \rightarrow \infty} x_n$ is

- (A) 1 for all α .
- (B) 1 or 0 depending on the value of α .
- (C) 1 or ∞ depending on the value of α .
- (D) 1, 0 or ∞ depending on the value of α .

Answer and Comments: (D). Denote the limit $\lim_{n \rightarrow \infty} x_n$ by L . When $\alpha = 1$, obviously $x_n = 1$ for all n and so $L = 1$. The limit when $\alpha = 1/2$ is also given as a standard example of limits. The most familiar proof is based on rationalisation. For in this case, we can write

$$\begin{aligned} x_n &= \sqrt{n+1} - \sqrt{n} \\ &= \frac{(n+1) - n}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n+1} + \sqrt{n}} \end{aligned} \quad (1)$$

As $n \rightarrow \infty$, both the terms in the denominator tend to ∞ and so $x_n \rightarrow 0$ as $n \rightarrow \infty$. So $L = 0$ in this case. This rules out the option (A). But it does not tell us how to handle the limit for other values of α , because rationalisation with other exponents such as $\frac{2}{3}$ is not easy while it is impossible when the exponent α is irrational.

But if we look at the real reason why the difference $\sqrt{n+1} - \sqrt{n}$ tends to 0 as n tends to infinity, we can get a clue. The difference between square roots of consecutive integers becomes very small for

large integers. Thus $\sqrt{5} - \sqrt{4} \approx 2.23 - 2 = 0.23$. But $\sqrt{10001} - \sqrt{10000}$ is extremely small because as compared with 10000, the addition of 1 is insignificant.

On the other hand, when the exponent is greater than 1, say $\alpha = 2$, the difference $(n + 1)^2 - n^2$ goes on widening with n and so in such cases $x_n \rightarrow \infty$ as $n \rightarrow \infty$.

Once this point is grasped, it is clear that $L = 0$ for $\alpha < 1$, $L = 1$ for $\alpha = 1$ and $L = \infty$ for $\alpha > 1$. So (D) is correct. This is not a rigorous proof. But it truly grasps the essence of it. And a question which rewards those who grasp such essence is a good question in an MCQ format. It is different from a question which permits a sneaky path.

For the sake of completeness, we include a proof based on the Lagrange's Mean Value Theorem. Consider the function $f(y) = y^\alpha$ defined for all $y > 0$. (We take the variable y instead of x so as to avoid confusion with x_n .) Then $f'(y) = \alpha y^{\alpha-1}$ for all $y > 0$. By LMVT, for every n there exists some $c_n \in (n, n + 1)$ such that

$$x_n = (n + 1)^\alpha - n^\alpha = f(n + 1) - f(n) = f'(c_n) = \alpha c_n^{\alpha-1} \quad (2)$$

We don't know exactly what c_n is and LMVT never tells you that. It is an existence theorem, i.e. it only guarantees the existence of something without giving an explicit formula for it. But no matter what c_n is, it is greater than n and so it tends to ∞ as $n \rightarrow \infty$. So, when $\alpha > 1$, the exponent $\alpha - 1$ is positive and $c_n^{\alpha-1}$ tends to ∞ . On the other hand, for $\alpha < 1$, the exponent $\alpha - 1$ is negative and as a result, $c_n^{\alpha-1}$ tends to 0 as $n \rightarrow \infty$.

A full length question which requires a candidate to give this (or some other) reasoning could have been asked. The interval for α is given as $(0, \frac{3}{2})$. Actually, $\frac{3}{2}$ has no role. Probably it is to discourage the special case $\alpha = 2$ where the limit is familiar.

Q.28 Let f be a continuous strictly increasing function from $[0, \infty)$ onto $[0, \infty)$ and $g = f^{-1}$ (that is, $f(x) = y$ if and only if $g(y) = x$). Let $a, b > 0$ and $a \geq b$. Then

$$\int_0^a f(x)dx + \int_0^b g(y)dy$$

is

- (A) greater than or equal to ab .
- (b) less than ab .
- (C) always equal to ab .
- (D) always equal to $\frac{af(a) + bg(b)}{2}$.

Answer and Comments: (A). In such problems, where there is no obvious clue, it pays to do a few special cases. The simplest example of a function $f(x)$ satisfying the given hypothesis is $f(x) = x$. This function is its own inverse and so for this function, $g(y) = y$. By a direct calculation, for any $a, b > 0$ we have

$$\int_0^a f(x)dx + \int_0^b g(y)dy = \frac{a^2}{2} + \frac{b^2}{2} \quad (1)$$

which is greater than or equal to ab by the A.M.-G.M. inequality. In fact, strict inequality holds when $a \neq b$. So options (B) and (C) are ruled out. (A) holds true. (In fact, since $a \neq b$, strict inequality holds.) Since $f(a) = a$ and $g(b) = b$ we see that (D) also holds.

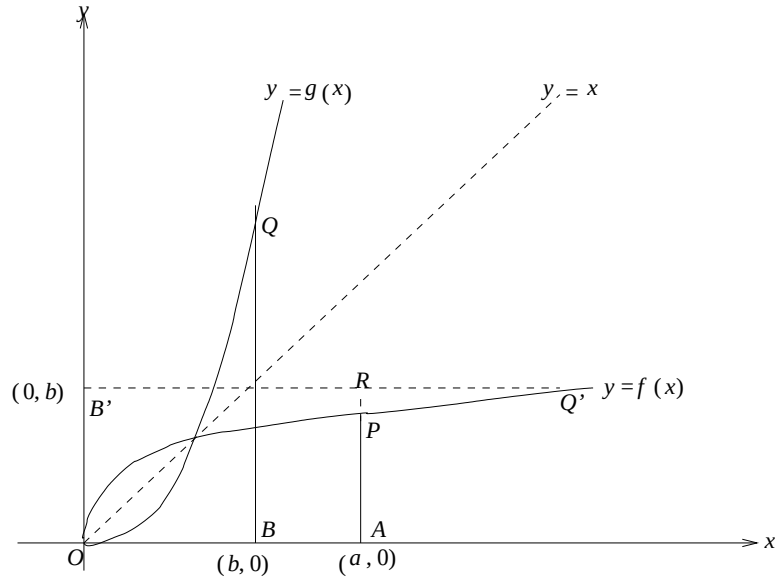
Let us now try, $f(x) = \sqrt{x}$ which is strictly increasing from $[0, \infty)$. Here the inverse function is $g(y) = y^2$ and so

$$\int_0^a f(x)dx + \int_0^b g(y)dy = \frac{2a^{3/2}}{3} + \frac{b^3}{3} \quad (2)$$

This time $\frac{af(a) + bg(b)}{2} = \frac{a^{3/2} + b^3}{2}$ which does not match with (2). So option (D) is also ruled out. Therefore, if at all one of the answers is correct, it has to be (A).

But that is far from complete. Before tackling the general case, let us check that (A) holds in this particular example. This can be done by applying the A.M.-G.M. inequality in a tricky way. We write the R.H.S. of (2) as the sum of *three* terms, viz. $\frac{a^{3/2}}{3} + \frac{a^{3/2}}{3} + \frac{b^3}{3}$. The A.M.-G.M. inequality now gives that this expression is at least $(a^{3/2}a^{3/2}b^3)^{1/3} = (a^3b^3)^{1/3} = ab$. (A similar trick was applied in the alternate solution to Q.24 given above.) So (A) holds here too.

Let us now show that (A) holds for every $f(x)$ which is a continuous, strictly decreasing function from $[0, \infty)$ to itself. The crucial property we need is that the graphs of $y = f(x)$ and $y = g(x)$ are reflections of each other in the line $y = x$ as shown in the figure below. (As familiar illustrations compare the graphs of $y = \sqrt{x}$ and $y = x^2$ or the graph of $y = e^x - 1$ with that of $y = \ln(x + 1)$.)



We make use of this symmetry as follows. Let $a, b > 0$ and let $A = (a, 0)$ and $B = (b, 0)$ be the points on the x -axis. In the integral $\int_0^b g(y)dy$, y is only a dummy variable and so the integral can also be written as $\int_0^b g(x)dx$. So our job is to prove that

$$\int_0^a f(x)dx + \int_0^b g(x)dx \geq ab \quad (3)$$

We shall do this geometrically. The first integral is the area of the region $OAPQ$ where $P = (a, f(a))$. Similarly, the second integral is the area of the region $OBQO$ where $Q = (b, g(b))$. By symmetry about the line $y = x$, this area equals that of the region $OB'Q'O$ where $B'=(0, b)$ and $Q' = (g(b), b)$. We assume, without loss of generality that

$b < a$. Then the union of the regions $OAP O$ and $OB'Q'O$ covers the rectangle $OARB'$ whose area is ab . So (3) holds.

It is not clear what is gained by stipulating $a \neq b$ when the option (A) allows possible equality. As already noted, the A.M.-G.M. inequality follows as a special case by taking $f(x) = x$. In that case both the regions $OAP O$ and $OB'Q'O$ are right angled equilateral triangles of sides a, b respectively. Their union covers the rectangle $OARB'$ whose area is ab . Thus we get a purely geometric proof of the A.M.-G.M. inequality, without the use of integrals. This geometric proof of the A.M.-G.M. inequality is far less well known than the algebraic proof based on the fact that the square of every real number is non-negative.

The inequality asked in this question is popularly called **Young's inequality**. Choosing the function $f(x)$ suitably it leads to several important special cases. For example, taking $f(x) = x^{p-1}$ we get

$$\frac{a^p}{p} + \frac{p-1}{p}b^{p(p-1)} \geq ab \quad (4)$$

which is often recast in a symmetric form as

$$\frac{a^p}{p} + \frac{b^q}{q} \geq ab \quad (5)$$

where $a, b > 0$ $p > 1$ and q is such that $\frac{1}{p} + \frac{1}{q} = 1$. (The A.M.-G.M. inequality results if we take $p = 2$.)

This question would be excellent as a full length question. In the MCQ format the correct answer can be guessed just by working out a couple of simple examples.

Q.29 The sum of the series $\sum_{n=1}^{\infty} n^2 e^{-n}$ is

$$(A) \frac{e^2}{(e-1)^3}. \quad (B) \frac{e^2 + e}{(e-1)^3}. \quad (C) \frac{3}{2}. \quad (D) \infty.$$

Answer and Comments: (B). There are very few infinite series that can be evaluated exactly. By far, the most well known among them is

the infinite geometric series

$$1 + x + x^2 + \dots + x^n + \dots = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \quad (1)$$

which is valid for all $x \in (-1, 1)$. It can be shown that in this interval term-by-term differentiation of this series is valid. As a result,

$$1 + 2x + 3x^2 + \dots + nx^{n-1} + \dots = \sum_{n=1}^{\infty} nx^{n-1} = \frac{d}{dx} \left(\frac{1}{1-x} \right) = \frac{1}{(1-x)^2} \quad (2)$$

Multiplying both the sides by x , we get

$$\sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2} \quad (3)$$

Differentiating term-by-term again, we get

$$\begin{aligned} \sum_{n=1}^{\infty} n^2 x^{n-1} &= \frac{d}{dx} \left(\frac{x}{(1-x)^2} \right) \\ &= \frac{1}{(1-x)^2} + \frac{2x}{(1-x)^3} \\ &= \frac{1+x}{(1-x)^3} \end{aligned} \quad (4)$$

Multiplying both the sides by x again,

$$\sum_{n=1}^{\infty} n^2 x^n = \frac{x(1+x)}{(1-x)^3} \quad (5)$$

This is valid for all $x \in (-1, 1)$. In particular, it is valid if we put $x = \frac{1}{e} = e^{-1}$. So we get

$$\begin{aligned} \sum_{n=1}^{\infty} n^2 e^{-n} &= \frac{\frac{1}{e} \times \frac{e+1}{e}}{\left(1 - \frac{1}{e}\right)^3} \\ &= \frac{e^2 + e}{(e-1)^3} \end{aligned} \quad (6)$$

Hence (B) is correct. The problem is straightforward for those who know about term-by-term differentiation of infinite power series. The

proof of the validity of such term-by-term differentiation is well beyond the Higher Secondary School. The problem can also be solved by resorting to differentiation of only a finite sum of functions, whose validity is established in elementary calculus. Thus we begin with

$$1 + x + x^2 + \dots + x^n = \sum_{k=0}^n x^k = \frac{1 - x^{n+1}}{1 - x} \quad (7)$$

If we call this sum as $S_n(x)$, it is the n -th (or the $(n+1)$ -th, depending upon the convention) partial sum of the infinite series $\sum_{k=0}^{\infty} x^k$. If we let $n \rightarrow \infty$, then on the R.H.S., $x^{n+1} \rightarrow 0$ if $|x| < 1$. So we get (1). If, instead, we differentiate both the sides of (7) w.r.t. x and multiply by x , we get

$$xS'_n(x) = \sum_{k=1}^n kx^k = x \times \frac{d}{dx} \left(\frac{1 - x^{n+1}}{1 - x} \right) \quad (8)$$

The R.H.S. comes out as $x \times \left[\frac{-(n+1)x^n(1-x) + (1-x^{n+1})}{(1-x)^2} \right]$. It can be simplified further. But we are more interested in its limit as $n \rightarrow \infty$. Here we need that for $|x| < 1$, not only x^{n+1} but $(n+1)x^n$ also tends to 0 as $n \rightarrow \infty$. So we get (3). One more differentiation and multiplication by x will give (5).

Yet another approach is to treat the series $\sum_{k=1}^n kx^k$ as an arithmetico-geometric series. The details are given in Comment No. 6 of Chapter 5. But obviously, assuming the validity of term-by-term differentiation for an infinite power series is the most efficient method, because the hard work is done by somebody else.

Q.30 Let $f : [0, 1] \rightarrow [-1, 1]$ be a non-zero function such that

$$f(2x) = 3f(x), \quad x \in [0, \frac{1}{2}].$$

Then, $\lim_{x \rightarrow 0^+} f(x)$ is equal to

$$(A) \frac{1}{2}. \quad (B) \frac{1}{3}. \quad (C) \frac{2}{3}. \quad (D) 0.$$

Answer and Comments: (D). The relation given to hold for the function $f(x)$ is called a **functional equation**. These are studied in Chapter 20. Sometimes questions about them have sneaky answers. For example, in the present problem, call the limit as L . As x tends to 0 from the right, the R.H.S. tends to $3L$. But as x tends to 0 from the right, so does $2x$. So the L.H.S. also tends to L . This gives the equation $L = 3L$ whence $L = 0$.

This solution presupposes that $\lim_{x \rightarrow 0^+} f(x)$ exists. What we have proved is that *in case this limit exists*, it must be 0. This is all right for answering an MCQ. But for a truly complete answer, we must show $f(x) \rightarrow 0$ as $x \rightarrow 0^+$ by a rigorous argument. We shall, in fact, do so by directly verifying the $\epsilon - \delta$ definition of a limit. So let $\epsilon > 0$ be given. We have to find some $\delta > 0$ such that

$$|f(x) - 0| < \epsilon \text{ whenever } 0 < x < \delta \quad (1)$$

The basic idea is as follows. Fix any $0 < x_0 < \frac{1}{2}$. The numbers $x_0, 2x_0, 4x_0, \dots, 2^r x_0, \dots$ are increasing and $2^r x_0 \rightarrow \infty$ as $r \rightarrow \infty$. So, there will be a unique positive integer r such that

$$2^r x_0 \in (1/2, 1] \quad (2)$$

For example, if $x_0 = \frac{3}{100}$, then $r = 5$ because $3 \times 2^5 = 96 \in (50, 100]$ and so $3^5 \times \frac{3}{100} \in (1/2, 1]$. Obviously r will increase as x_0 moves closer to 0. Now repeated applications of the given relation $f(2x) = 3f(x)$ give $f(2x_0) = 3f(x_0), f(4x_0) = \frac{f(x_0)}{9}, \dots$, till finally,

$$f(2^r x_0) = 3^r f(x_0) \quad (3)$$

We cannot go further because the relation is given to hold only for $x \in [0, \frac{1}{2}]$. We are not given the values of $f(x)$ anywhere. But we are given that $|f(x)| \leq 1$ for all $x \in [0, 1]$. (The number 1 is not crucial. Any upper bound on $|f(x)|$ would work.)

From this upper bound and (3) we get

$$|f(x_0)| \leq \frac{1}{3^r} \quad (4)$$

If we have some positive number x_1 smaller than x_0 , the corresponding integer, say s may be higher than r . Still we shall have

$$|f(x_1)| \leq \frac{1}{3^s} \leq \frac{1}{3^r} \quad (5)$$

Thus we see that if we fix $x_0 > 0$ and the r corresponding to it, then we shall have

$$|f(x)| \leq \frac{1}{3^r}, \text{ whenever } 0 < x < x_0 \quad (6)$$

Now, given $\epsilon > 0$, there exists some m such that

$$\frac{1}{3^m} < \epsilon \quad (7)$$

(This follows because $3^m \rightarrow \infty$ as $m \rightarrow \infty$.) It is now easy to find a desired δ . We merely take it as $\frac{1}{2^m}$. The r that works for this δ is m . Taking $x_0 = \delta$, (1) follows from (6) and (7).

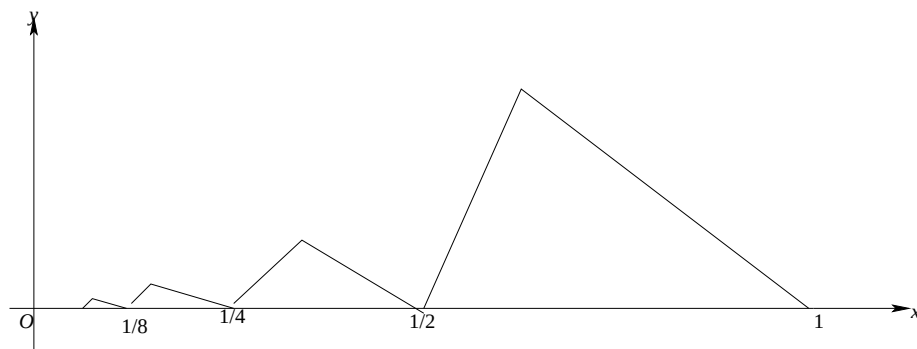
The argument can be given a little more systematically in terms of logarithms. Specifically, for a fixed $x_0 \in (0, 1/2)$ the integer r we defined above is simply the integral part of the logarithm of x_0 w.r.t. the base $\frac{1}{2}$. But the essential idea remains the same. We can also argue that the upper bound 1 on $|f(x)|$ for $x \in (1/2, 1]$ implies an upper bound $\frac{1}{3}$ on $|f(x)|$ for $x \in (1/4, 1/2]$, an upper bound $\frac{1}{9}$ for $x \in (1/8, 1/4]$ and, by induction, an upper bound $\frac{1}{3^r}$ on $|f(x)|$ for $x \in (\frac{1}{2^{r+1}}, \frac{1}{2^r}]$. In fact this gives a more visual proof that $\lim_{x \rightarrow 0^+} f(x)$ exists and equals 0.

Yet another problem which is suitable as a full length question. To make it suitable as an MCQ, one of the options should have been that the limit $\lim_{x \rightarrow 0^+} f(x)$ does not exist.

Although the problem does not give any particular function which satisfies the given functional equation, a question that arises naturally is whether such a function indeed exists, as otherwise the problem would be vacuous. To construct such an example, note that the given functional equation is of the same spirit as periodicity. But, for a genuinely periodic function, the periods as well as the amplitudes are uniform. In our case, as you go from 1 to 0, the lengths of the periods

form a G.P. with common ratio $\frac{1}{2}$ while the amplitudes form a G.P. with common ratio $\frac{1}{3}$.

Once this point is understood, it is easy to construct such a function using the same functional equation. Define $f(x)$ arbitrarily for $x \in (1/2, 1]$. Then we have no choice in defining it for $x \in (1/4, 1/2]$. We have to let $f(x) = \frac{f(2x)}{3}$. We then define $f(x)$ uniquely for $x \in (1/8, 1/4]$. If we want to make f continuous, we better ensure that $f(1) = 3 \lim_{x \rightarrow \frac{1}{2}^-} f(x)$. The figure below shows the graph of one such function. (It is only a qualitative sketch. The slopes of the segments form a G.P. with common ratio $\frac{2}{3}$ as you go from 1 to 0.)



ISI BStat-BMath-UGB-2016 Paper with Comments

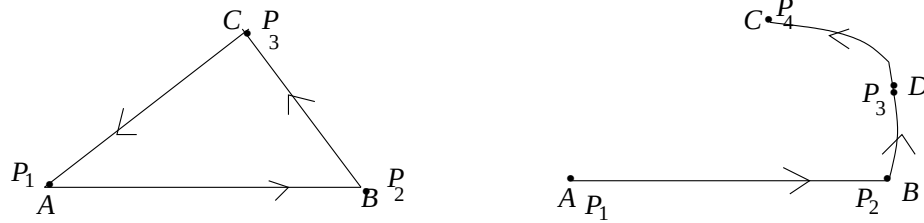
- Q 1.** Suppose that in a sports tournament featuring n players, each pair plays one game and there is always a winner and a loser (no draws). Show that the players can be arranged in an order $P_1, P_2, \dots, P_n$ such that player P_i has beaten P_{i+1} for all $i = 1, 2, \dots, n$.

Solution: Note that it is *not* given that whenever a player A beats B and B beats C , then A beats C . In other words, the relation ‘beating’ is not given to be transitive. If it were, the problem would be very easy. We first show that there is some player who is the champion, i.e. who has beaten every other player. If this is not the case, then there will be a chain of players $Q_1 - Q_2 - \dots$ in which every Q_i loses to Q_{i+1} and

because of transitivity no two players in this chain can be the same. But this is impossible as there are only finitely many players.

When the relation of beating is transitive, we simply let P_1 be the champion, P_2 be the champion among all players excluding P_1 and so on. Indeed this is the only way they can be arranged.

The problem becomes interesting only when the relation of beating is not transitive. In such a case, there will be three players A, B, C such that A beats B , B beats C and C beats A . Such a situation is suggestively called a **directed cycle** of length 3 and pictured in (a) of the figure below where the players are represented by vertices and a directed edge (which does not have to be a straight line) between two players indicates that the player at its start beats the one at the end.



(a) a 3-cycle

(b) insertion of a new vertex

When we have three players and they form a cycle of length 3 (or a 3-cycle for short) as in (a) it does not matter which player we put first. So we let $P_1 = A$, $P_2 = B$ and $P_3 = C$ and form the chain $P_1 - P_2 - P_3$. Here P_3 beats P_1 . But that does not violate the condition since it is to hold only for two consecutively placed players.

We want to prove the assertion by induction on n . So, let us see how a fourth player, say D will fit into the chain $P_1 - P_2 - P_3$ we have formed. We compare D with P_3 . If P_3 beats D , we call D as P_4 . Suppose, however, that D beats P_3 . Then we compare D with P_2 . If P_2 beats D we insert D between P_2 and P_3 so that our new chain will now be $P_1 - P_2 - D - P_3$ as shown in(b) above. We can now rename D as P_3 and old P_3 (i.e. C) as P_4 . Then the chain $P_1 - P_2 - P_3 - P_4$ satisfies the condition. In case D beats P_2 too, but not P_1 , we break the edge from P_1 to P_2 and insert D between P_1 and P_2 so that the new chain now is $P_1 - D - P_2 - P_3$ which we may relabel as $P_1 - P_2 - P_3 - P_4$.

Finally, if D beats P_1 too (in addition to P_3 and P_2), then we put D right at the start of the chain.

It is now clear how the proof proceeds by induction on n . For $n = 2$, we have no choice in ordering the players. Suppose that we have n players where $n > 2$ and $n - 1$ of these have already been arranged in a chain $P_1 - P_2 - \dots - P_{n-1}$ so that P_i beats P_{i+1} for $i = 1, 2, \dots, n - 2$. We take the n -th player Q and start comparing him/her with $P_{n-1}, P_{n-2}, \dots$ successively till we encounter some i such that Q beats P_i but not P_{i-1} . Then we insert the new player between P_{i-1} and P_i , call P_j as P_{j+1} for $j = i, i + 1, \dots, n - 1$ and finally call Q as P_i . If we do not encounter such an adjacent pair, it means that the new player beats all. In that case we put him/her at the start of the chain. This completes the inductive step and the proof. (Actually, this proof works equally well even when there are no cycles.)

By picturing the players as vertices, putting an edge between every two players and directing it we have drawn what is called a **complete directed graph** on n vertices. In general, a directed graph consists of some vertices only some of which are joined by edges. We can also consider graphs whose edges are not directed. In a complete graph, every pair is joined by an edge and so it has $\binom{n}{2}$ edges where n is the number of vertices. The theory of graphs is visually appealing and has many practical applications. The result we have proved here says that in every complete directed graph, there is a directed path containing all vertices.

Q. 2 Consider a cubic polynomial $p(x) = ax^3 + bx^2 + cx + d$ where a, b, c, d are integers such that ad is odd and bc is even. Prove that not all roots of $p(x)$ can be rational.

Solution: We first observe a very simple fact about the rational roots of $p(x)$. Suppose $\frac{u}{v}$ is one such root in its reduced form, i.e. where the integers u, v have no common factor other than ± 1 . By a direct substitution and multiplying by v^3 gives

$$au^3 + bu^2v + cuv^2 + dv^3 = 0 \quad (1)$$

The integer u divides the first three terms and hence divides $dv^3 =$

$-au^3 - bu^2v - cuv^2$. But u is relatively prime to v and so u must divide d . By a similar reasoning v must divide a . (This elementary result is called the **rational root test**. It holds for any polynomial with integer coefficients and is useful in finding all possible rational roots of it. In particular, we see that the polynomials $x^2 - 2$ and $x^3 - 2$ have no rational roots, thereby proving that the square root as well as the cube root of 2 is irrational. See Exercises (10.11) and (10.11).)

Returning to the solution, since ad is odd, both a and d are odd. So, if $\frac{u}{v}$ is a rational root of $p(x)$ in a reduced form, then u, v are both odd. Now suppose that all three roots of $p(x)$ are rational, say, $\frac{u_1}{v_1}, \frac{u_2}{v_2}$ and $\frac{u_3}{v_3}$ all in their reduced forms. From the general formula for the sum of the roots of a cubic, we have

$$\frac{u_1}{v_1} + \frac{u_2}{v_2} + \frac{u_3}{v_3} = -\frac{b}{a} \quad (2)$$

Clearing the denominators, this becomes

$$a(u_1v_2v_3 + u_2v_3v_1 + u_3v_1v_2) = -b(v_1v_2v_3) \quad (3)$$

We already proved that the integers $u_1, u_2, u_3, v_1, v_2, v_3$ are all odd. Also a is odd. So the L.H.S. is an odd integer. This forces b to be odd too. The other relation the three roots of a cubic gives

$$\frac{u_1u_2}{v_1v_2} + \frac{u_2u_3}{v_2v_3} + \frac{u_3u_1}{v_3v_1} = \frac{c}{a} \quad (4)$$

which, upon simplification, gives

$$a(u_1u_2v_3 + u_2u_3v_1 + u_3u_1v_2) = c(v_1v_2v_3) \quad (5)$$

Once again the L.H.S. is odd. But then c is odd. As b, c are both odd, we get that bc is odd, contradicting the hypothesis. Hence not all three roots can be rational.

This solution is a simple application of the rational root test. But most of the time we are using only that a product of odd numbers is odd and the sum of an odd number of odd summands is odd.

Q. 3 Given the polynomial

$$f(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_n$$

with real coefficients, and $a_1^2 < a_2$, show that not all roots of $f(x)$ can be real.

Solution: Suppose the n roots of $f(x)$ (counted according to multiplicities) are $r_1, r_2, \dots, r_n$. Then we have a factorisation

$$f(x) = (x - r_1)(x - r_2) \dots (x - r_n) \quad (1)$$

Comparison of the like powers gives

$$a_1 = -(r_1 + r_2 + \dots + r_n) = -\sum r_i \quad (2)$$

$$a_2 = \sum_{i < j} r_i r_j \quad (3)$$

$$a_3 = -\left(\sum_{i < j < k} r_i r_j r_k\right) \quad (4)$$

$$\vdots \quad \vdots \quad (5)$$

$$a_n = (-1)^n r_1 r_2 \dots r_n \quad (6)$$

(The functions appearing on the right hand sides are called the **elementary symmetric functions** of the roots. Whenever a Σ notation is used it is understood that it means the sum of all similar terms. For example, in (3), the R.H.S. consists of the sum of all $\binom{n}{2}$ terms of the form $r_i r_j$ with $i < j$.)

For the solution we need only the first two of these relations. Squaring (2), we have

$$a_1^2 - a_2 = \left(\sum_i r_i\right)^2 - \sum_{i < j} r_i r_j = \sum_i r_i^2 + \sum_{i < j} r_i r_j \quad (7)$$

The R.H.S. can be written as $\frac{1}{2} \left[\left(\sum_i r_i\right)^2 + \sum_i r_i^2 \right]$ which is non-negative if all r_i 's are real. But this contradicts the hypothesis that $a_1^2 < a_2$. So not all the roots of $f(x)$ can be real.

The problem is very similar to Exercise (16.20)(a), where the condition given is $(n-1)a_1^2 < 2na_2$ instead of $a_1^2 < a_2$. The proof is almost the same. The solution to the last problem was also based on the elementary symmetric functions of the roots of a cubic. One of the solutions to Q.4 of Paper 1 was also based on it. So too many problems in a single test are based on the same idea.

There is also a more sophisticated solution using calculus. Repeated applications of the Rolle's theorem give that if a polynomial $f(x)$ has n real roots counted according to their multiplicities, then $f'(x)$ has $n-1$ real roots, $f''(x)$ has $n-2$ real roots and so on. Continuing, we get that $f^{(n-2)}(x)$ has 2 real roots. By a direct calculation, $f^{(n-2)}(x)$ is the quadratic polynomial $\frac{n!}{2}x^2 + (n-1)!a_1x + (n-2)!a_2$ whose roots are the same as those of the quadratic $\frac{n(n-1)}{2}x^2 + (n-1)a_1x + a_2$. But then the discriminant criterion gives $(n-1)^2a_1^2 \geq 2n(n-1)a_2$ and hence $a_1^2 \geq \frac{2n}{n-1}a_2$ which contradicts the hypothesis because $\frac{2n}{n-1} \geq 1$.

- Q. 4** Let d be a positive integer. Prove that there exists a right-angled triangle with rational sides and area equal to d if and only if there exists an arithmetic progression x^2, y^2, z^2 of squares of rational numbers whose common difference is d .

Solution: For the direct implication suppose that d is the area of a right angled triangle with sides a, b and $\sqrt{a^2 + b^2}$ all of which are rational. Then $d = \frac{1}{2}ab$ and by a direct calculation,

$$\left(\frac{a+b}{2}\right)^2 - \left(\frac{\sqrt{a^2+b^2}}{2}\right)^2 = \frac{a^2+b^2+2ab}{4} - \frac{a^2+b^2}{4} = \frac{2ab}{4} = d \quad (1)$$

By another such calculation, we have

$$\left(\frac{\sqrt{a^2+b^2}}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2 = \frac{a^2+b^2}{4} - \frac{a^2+b^2-2ab}{4} = \frac{2ab}{4} = d \quad (2)$$

So if we let $z = \frac{a+b}{2}$, $y = \frac{\sqrt{a^2+b^2}}{2}$ and $x = \frac{a-b}{2}$, we get a desired progression because a, b and $\sqrt{a^2+b^2}$ are all given to be rational.

The proof of the converse is equally easy. Assume that x, y, z are positive rationals such that

$$y^2 - x^2 = d = z^2 - y^2 \quad (3)$$

We let $a = z + x$, $b = z - x$ and $c = 2y$. Then a, b, c are all rational and by a direct calculation,

$$a^2 + b^2 = 2z^2 + 2y^2 = 4y^2 = c^2 \quad (4)$$

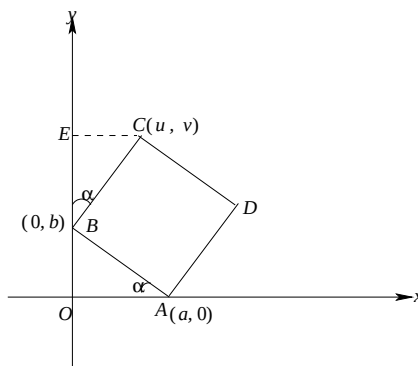
Therefore a, b, c form a right angled triangle with rational sides. As for its area, it is $\frac{1}{2}ab$ which equals $\frac{1}{2}(z+x)(z-x) = \frac{1}{2}(z^2 - x^2) = d$ by (3).

The solution nowhere requires that d is an integer. So this stipulation is redundant. In fact, it is likely to mislead a candidate into thinking that the problem has something to do with what are called **Pythagorean triples**, i.e. triples of integers which form the sides of a right angled triangle. All such triples can be obtained in a certain form, see Exercise (4.22). They are of the form $(2uv, u^2 - v^2, u^2 + v^2)$ for some positive integers u, v with $u > v$. (The simplest triple $(4, 3, 5)$ results if $u = 2, v = 1$.) Then $d = uv(u^2 - v^2)$. The factorisation of d as $(u^2 + uv)(uv - v^2)$ expresses d as $z^2 - y^2$ where $z = \frac{u^2 + uv}{2} + \frac{uv - v^2}{2} = \frac{u^2 + 2uv - v^2}{2}$ and $y = \frac{u^2 + uv}{2} - \frac{uv - v^2}{2} = \frac{u^2 + v^2}{2}$. Similarly the factorisation of d as $(u^2 - uv)(uv + v^2)$ expresses it as $y^2 - x^2$ where $x = \frac{u^2 - 2uv - v^2}{2}$. This gives the desired progression whose common difference is d .

But all this has little to do with the present problem. All that we need here is a manipulation of squares of the sum and difference of two numbers. A very disappointing problem. But perhaps today there is some place even for such problems, because those who are thoroughly brain washed by MCQ's may not even understand what is to be done in an 'if and only if' question.

- Q. 5.** Let $ABCD$ be a square two of whose adjacent vertices, say A, B , are on the positive X -axis and Y -axis, respectively. If C has co-ordinates (u, v) in the first quadrant, determine the area of $ABCD$ in terms of u and v .

Solution: Another straightforward problem. In such problems, once you draw the right diagram, the rest of the work is routine. Such a diagram is shown below.



Let $A = (a, 0)$ and $B = (0, b)$. If we can express b in terms of u and v , then we would know the side BC and hence also the area of the square $ABCD$ in terms of u and v . So the real task is to get hold of b . To do this draw a perpendicular CE from C to the Y -axis. Then $EC = u$. Now consider the right angled triangles OAB and EBC . Their longest sides AB and BC are equal, since $ABCD$ is a square. Also $BC \perp AB$ and $AO \perp BE$. Therefore $\angle OAB = \angle EBC$. As a result, the triangles $\triangle OAB$ and $\triangle EBC$ are congruent. Therefore $b = OB = EC = u$.

The rest is pure algebra. The point B now is $(0, u)$. By the distance formula

$$BC = \sqrt{(u - 0)^2 + (v - u)^2} = \sqrt{2u^2 + v^2 - 2uv} \quad (1)$$

(We can also calculate BC directly using the Pythagoras theorem. But anyway the distance formula is also based on the Pythagoras theorem.) Therefore, the area of the square $ABCD$ is simply $(BC)^2 = 2u^2 + v^2 - 2uv$. (Yet another approach is to view the point $C = (u, v)$ as a complex number $z = u + iv$, with $A = a$ and $B = ib$. Since BC is obtained by rotating BA anticlockwise through 90 degrees, we get $z - ib = i(a - ib) = b + ia$. Equating the real parts of the two sides gives $u = b$.)

As a geometry problem, this problem comes nowhere close to the many challenging problems in the golden days of JEE when pure geometry was a part of its syllabus. All it requires is the construction of dropping a perpendicular CE and then proving that $\triangle OAB \cong \triangle EBC$. But pure geometry has long been abolished from JEE and is given only a lip service in the school curricula. So chances are that today even

this task might be challenging to many students. Those who draw a correct diagram will probably realise that $b = u$ and then give the correct answer in an MCQ. But it is doubtful how many of them will be able to give the correct reasoning for it. Probably, this problem has been designed to test this fast declining quality on the part of the candidates. So it deserves to be commended even though as a geometry problem it is far from challenging.

- Q. 6.** Let a, b, c be the sides of a triangle and A, B, C be the angles opposite to these sides respectively. If

$$\sin(A - B) = \frac{a}{a + b} \sin A \cos B - \frac{b}{a + b} \cos A \sin B,$$

then prove that the triangle is isosceles.

Solution: Although the problem does not say which of the two angles from A, B, C are to be shown equal, it is obvious that we must show $A = B$. The dependence on the sides a, b in the data can be eliminated by observing that the sides are always proportional to the sines of the angles opposite to them. So multiplying both the sides of the given relationship by $a + b$ and replacing a, b by $\sin A$ and $\sin B$ respectively, the given condition becomes

$$(\sin A + \sin B) \sin(A - B) = \sin^2 A \cos B - \sin^2 B \cos A \quad (1)$$

Expanding $\sin(A - B)$ this becomes

$$(\sin A + \sin B)(\sin A \cos B - \cos A \sin B) = \sin^2 A \cos B - \sin^2 B \cos A \quad (2)$$

Expanding the L.H.S., two of the terms cancel with the R.H.S. and we get

$$\sin A \sin B \cos B = \sin A \cos A \sin B \quad (3)$$

Cancelling $\sin A$ and $\sin B$ (which are always positive in any triangle) we get

$$\cos A = \cos B \quad (4)$$

whence $A = B$ because the cosine function is one-to-one on the interval $(0, \pi)$ which contains both A and B . ■

Once again, this problem ranks nowhere close to any of the interesting trigonometric problems asked in the JEE way back when it had a subjective component. But like the last problem, it deserves a welcome. There is no way a question like this can be asked as an MCQ. If we artificially convert it to an MCQ by giving the four options as, say (i) $A = B$, (ii) $A = 2B$, (iii) $B = 2A$ and (iv) none of these necessarily, then nearly all candidates will pick (i) without batting an eyelid. But many of them may not be able to give the simple solution above. And an even more alarming possibility is that some of them will think that (i) is true because when it holds the given condition is satisfied! In other words, they cannot distinguish an implication statement from its converse. This is the abyss to which the reasoning ability of many students has been reduced by the exclusive dependence on MCQ's in all important examinations. So just as even one pail of water in a desert means a lot, a question like this has a place in today's educational scenario.

Q. 7 Let f be a differentiable function such that $f(f(x)) = x$ for all $x \in [0, 1]$. Suppose $f(0) = 1$. Determine the value of $\int_0^1 (x - f(x))^{2016} dx$.

Solution: As in Q.12 of Paper 1, the exponent 2016 is apparently chosen to conform to a recent fashion. But there it had at least some role because its factorisation was crucial to getting the solution. In this problem it is a mere show piece. The real task is to determine the function $f(x)$ from the data. We are given that $f(f(x)) = x$ for all $x \in [0, 1]$. This means that f is a bijection from $[0, 1]$ onto itself and further that it is its own inverse. Being continuous and one-to-one, f has to be either strictly increasing or strictly decreasing. (This is a consequence of the Intermediate Value Property, see Comment No. 6 of Chapter 16.) Since $f(0) = 1$, the second possibility must hold and we must have $f(1) = 0$. So f is a function from $[0, 1]$ onto itself which maps it upside down. The most natural example of such a function that comes to the mind is the reflection into the midpoint $\frac{1}{2}$. The formula for $f(x)$ then is

$$f(x) = 1 - x \tag{1}$$

If this guess is correct, then the given integral, say I , can be evaluated

as follows.

$$\begin{aligned}
I &= \int_0^1 (x - f(x))^{2016} dx = \int_0^1 (2x - 1)^{2016} dx \\
&= \frac{(2x - 1)^{2017}}{4034} \Big|_0^1 = \frac{1 - (-1)}{4034} = \frac{1}{2017}
\end{aligned} \tag{2}$$

In an MCQ, this will probably pass as a correct answer. But now the ball game is different. We have to give a rigorous proof that our guess (1) is correct. Unfortunately, it turns out that $f(x) = 1 - x$ is not the only function from $[0, 1]$ to itself which satisfies the conditions $f(f(x)) = x$ for all $x \in [0, 1]$ with $f(0) = 1$.

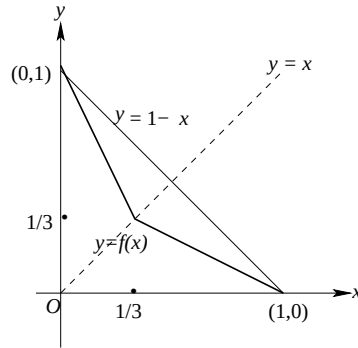
Let us see how another such example can be constructed. (1) was obtained as the reflection into the point $\frac{1}{2}$, which is the mid-point of the interval $[0, 1]$. Instead, we can take the reflection into some other point, say $\frac{1}{3}$. But now to stay within the interval $[0, 1]$, the reflection of one part will have to be multiplied by the ratio of the length of the second part to that of the first. Thus, suppose $0 \leq x \leq 1/3$. Then x is at a distance $\frac{1}{3} - x$ from $1/3$ (the centre of the reflection) and on its left. So we let $f(x)$ be the point on the right of $1/3$ and at a distance $2(\frac{1}{3} - x)$ from it. This gives

$$\frac{1}{3} + 2(\frac{1}{3} - x) = 1 - 2x \tag{3}$$

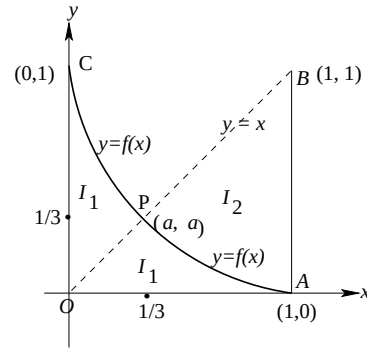
for all $[0, 1/3]$. By an analogous reasoning, we get $f(x) = \frac{1}{3} - \frac{1}{2}(x - \frac{1}{3})$ for all $x \in [1/3, 1]$. Clubbing together, we get a function $f(x)$

$$f(x) = \begin{cases} -2x + 1, & \text{if } 0 \leq x \leq \frac{1}{3} \\ -\frac{1}{2}x + \frac{1}{2}, & \text{if } \frac{1}{3} \leq x \leq 1 \end{cases} \tag{4}$$

The graph of $f(x)$ is shown in (a) of the figure below (along with the graph of $y = 1 - x$ for the sake of comparison).



(a)



(b)

It is clear from the graph that $f(0) = 1$. The verification that $f(f(x)) = x$, or in other words, that f is its own inverse can be done by appropriate substitutions into (4). But the best way to see this is that the graph of $y = f(x)$ is symmetric about the line $y = x$. (The solution to Q.28 of Paper 1 was based crucially on the fact that if the functions f and g are inverses of each other then the graphs of $y = f(x)$ and $y = g(x)$ are reflections of each other in the line $y = x$. In the present problem g equals f . The function f is not differentiable at $\frac{1}{3}$. But non-differentiability at a single point causes no problems in integration. At any rate, we can take $f(x)$ to be the function whose graph is an arc of a circle passing through the points $(0, 1)$, $(\frac{1}{3}, \frac{1}{3})$ and $(1, 0)$ as shown in (b).)

Let us calculate I for the function $f(x)$ defined by (4). Here we have to split the interval of integration at the point $\frac{1}{3}$. We then get

$$\begin{aligned}
 I &= \int_0^{1/3} (3x - 1)^{2016} dx + \int_{1/3}^1 \left(\frac{3}{2}x - \frac{1}{2}\right)^{2016} dx \\
 &= \frac{1}{3} \frac{(3x - 1)^{2017}}{2017} \Big|_0^{1/3} + \frac{2}{3} \frac{\left(\frac{3}{2}x - \frac{1}{2}\right)^{2017}}{2017} \Big|_{1/3}^1 \\
 &= \frac{1}{3} \times \frac{0 - (-1)}{2017} + \frac{2}{3} \times \frac{1 - 0}{2017} = \frac{1}{2017}
 \end{aligned} \tag{5}$$

If we take any $a \in (0, 1)$ and define f as the reflection into the point (a, a) (scaled suitably so that 0 and 1 are taken to 1 and 0 respectively), then the calculations will be very similar and we shall get that

$$I = a \times \frac{1}{2017} + (1 - a) \times \frac{1}{2017} = \frac{1}{2017} \tag{6}$$

This suggests that no matter which $f(x)$ we take, as long as it satisfies the conditions in the problem, the integral $I = \int_0^1 (x - f(x))^{2016} dx$ will always equal $\frac{1}{2017}$. But how do we prove it when the function $f(x)$ is not given by any explicit formula and all we know about it is that it is differentiable, is its own inverse and $f(0) = 1$? This is quite tricky. The trick is that with a suitable substitution, we convert I to another definite integral J . Neither I nor J can be evaluated easily. But sometimes the sum $I + J$ can. (See Comment No. 14 of Chapter 18 for illustrations. In case this sounds paradoxical, recall the trick in evaluating the sum, say $S_n = 1 + 2 + \dots + n$ where n is a positive integer. Simply write the sum backwards as $S_n = n + (n - 1) + \dots + 2 + 1$. If we add the two expressions for S_n term by term, we get $S_n = (1 + n) + (2 + n - 1) + (3 + n - 2) + \dots + (n - 1 + 2) + (n + 1) = n(n + 1)$ which gives S_n as $\frac{n(n + 1)}{2}$. Two jaggy things can sometimes fit together to give something smooth as in a jigsaw puzzle.)

So, once again, let $I = \int_0^1 (x - f(x))^{2016} dx$. Since f is its own inverse, we can use the substitution $x = f(u)$ which transforms the interval $[0, 1]$ bijectively to the interval $[0, 1]$ but traced backwards because $f(0) = 1$ and $f(1) = 0$. As f is differentiable, we get

$$I = \int_1^0 (u - f(u))^{2016} f'(u) du \quad (7)$$

which, in a way, is more complicated than I . But reversing the upper and the lower limits of integration, we get

$$I = - \int_0^1 (u - f(u))^{2016} f'(u) du \quad (8)$$

Adding this to the original expression for I (after changing the dummy variable of integration x to u) we have

$$2I = \int_0^1 (u - f(u))^{2016} (1 - f'(u)) du \quad (9)$$

This is something we can evaluate using the fundamental theorem of calculus because now the integrand is the derivative of $\frac{(u - f(u))^{2017}}{2017}$

w.r.t. u . So

$$\begin{aligned}
 2I &= \left. \frac{(u - f(u))^{2017}}{2017} \right|_{u=0}^{u=1} \\
 &= \frac{(1 - f(1))^{2017} - (0 - f(0))^{2017}}{2017} \\
 &= \frac{1 - (-1)}{2017} = \frac{2}{2017}
 \end{aligned} \tag{10}$$

So, finally, $I = \frac{1}{2017}$.

This problem is an excellent example of a good question which would be ruined if asked as an MCQ because then the correct answer can be arrived simply by working out the simplest case $f(x) = 1 - x$. Admittedly the trick here is of limited applicability. It works only because the integrand is an even power of $x - f(x)$. But that is the name of the game. With an odd exponent, the integral will not be independent of f .

The trick above will work, more generally, for an integral of the form $\int_0^1 h(x - f(x))dx$ where $h(t)$ is an even function of t . Suppose $h(t) = |t|$. Then I would be $\int_0^1 |x - f(x)|dx$. In this case the trick above will give $I = \frac{1}{2}$. We can also show this geometrically using the fact that the graph of $y = f(x)$ is symmetric w.r.t. the line $y = x$. Note first that because f is strictly decreasing with $f(0) = 1$ and $f(1) = 0$, the graph will cut the line $y = x$ at a unique point of the form (a, a) for some $a \in (0, 1)$. This is a consequence of the Intermediate Value Property applied to the function $f(x) - x$. See (b) of the figure above. (Earlier we took $a = 1/3$. But that was only an illustration.) Call the points $(1, 0)$, $(1, 1)$, $(0, 1)$ and (a, a) as A, B, C, P respectively. Now divide the integral I as $I_1 + I_2$ where

$$\begin{aligned}
 I_1 &= \int_0^a |x - f(x)|dx \\
 \text{and } I_2 &= \int_a^1 |x - f(x)|dx
 \end{aligned} \tag{11}$$

Then I_1 and I_2 are, respectively, the areas of the regions $O - P - C - O$ and $A - B - P - A$. Because of symmetry of the graph about the line

$y = x$, I_1 is also the area of the region $O - A - P - O$. Hence $I_1 + I_2$ is the area of the triangle with vertices O, A, B . So its value is $\frac{1}{2}$.

A similar geometric approach to evaluate $\int_0^1 (x - f(x))^{2016} dx$ appears complicated.

Although the problem involves a trick which might elude some good candidates, some partial credit can be given if the candidate has worked out the simplest case, viz. $f(x) = 1 - x$. If a candidate has done one more example such as the function in (4), the partial credit can be increased. This is the advantage of a subjective test. It does not classify candidates only as black and white.

Q. 8. Let $\{a_n\}_{n \geq 1}$ be a sequence of real numbers defined recursively by

$$a_{n+1} = \frac{3a_n}{2 + a_n},$$

for all $n \geq 1$.

- (i) If $0 < a_1 < 1$, then prove that $\{a_n\}_{n \geq 1}$ is increasing and $\lim_{n \rightarrow \infty} a_n = 1$.
- (ii) If $a_1 > 1$, then prove that $\{a_n\}_{n \geq 1}$ is decreasing and $\lim_{n \rightarrow \infty} a_n = 1$.

Solution: As is to be expected, when a sequence is recursively defined, mathematical induction is the stock method for proving any assertion about its terms. The adjectives ‘increasing’ and ‘decreasing’ when used in connection with real valued functions (which includes sequences too, as they are certain functions) often invite some controversy because of a difference of conventions. According to one convention, ‘increasing’ means ‘strictly increasing’ while the other convention allows possible equality too. If equality is to be allowed, the first convention uses the non-committal ‘non-increasing’ or ‘monotonically increasing’. The first convention is getting more acceptance possibly because the terms ‘non-negative’ and ‘positive’ are both in use. Well written books clearly specify which convention is adopted. The same holds for classroom tests conducted by instructors of courses. But in the various entrance examinations, where there is no other channel of communication between the candidates and the paper setters, it is a good idea to specify the convention. Or else it has to be inferred from the context.

In the present problem, this distinction does not matter. We are given that $a_{n+1} = \frac{3a_n}{2+a_n}$ and the only way a_{n+1} will equal a_n is when $a_n = 1$ or 0 . But even if this happens for one value of n , it would imply that $a_1 = 1$ or 0 correspondingly. This possibility is excluded both in (i) and (ii). So, in this problem, we shall take ‘increasing’ to mean ‘strictly increasing’ with a similar interpretation for ‘decreasing’.

In (i) we have to prove that $a_{n+1} > a_n$ for every n . Let us begin with $n = 1$. By definition $a_2 = \frac{3a_1}{2+a_1}$. As $a_1 > 0$, proving $a_2 > a_1$ amounts to showing that $3 > 2 + a_1$ which is true since it is also given that $a_1 < 1$. So there is no difficulty in starting the induction.

Coming to the inductive step, assume that we have proved that

$$a_n > a_{n-1} > a_{n-2} > \dots > a_2 > a_1 \quad (1)$$

and now want to prove that $a_{n+1} > a_n$, or equivalently,

$$\frac{3a_n}{2+a_n} > a_n \quad (2)$$

Note that by (1), $a_n > 0$ and so (2) reduces to showing

$$3 > 2 + a_n \quad (3)$$

For this we need $a_n < 1$. But that is not given to hold except for $n = 1$. Nor does it follow automatically from (1). So, even though (i) does not ask it, we shall prove by induction the stronger statement

$$P(n) \equiv a_n < a_{n+1} \text{ and } a_n < 1 \quad (4)$$

for every $n \geq 1$. We begin by proving $P(1)$. We already proved that $a_2 > a_1$. But now we also have to prove that $a_2 < 1$. This is equivalent to showing that $3a_1 < 2 + a_1$, i.e. $2a_1 < 2$. But that is true because we are given $a_1 < 1$. (We also need here that 2 is positive. But such incidental details are not expected from the candidates. In a subjective examination where a candidate has to give reasoning, the degree of completeness of justifications is subjective. But surely nobody expects a candidate to say ‘by commutativity of multiplication’ every time it is used. Although no explicit rules are set anywhere, when subjective

examinations become the norm, a tacit understanding evolves as to which steps in the argument need to be justified. The time allotted to a question also plays a role here.)

Now, coming to the inductive step, assume that for some integer n we have proved

$$1 > a_n > a_{n-1} > a_{n-2} > \dots > a_2 > a_1 \quad (5)$$

and now want to prove that $a_{n+1} > a_n$ and also that $a_{n+1} < 1$. Now that we know that $a_n < 1$, the earlier proof that $a_{n+1} > a_n$ has no hitch. But now the work is also more. That is, we must prove $a_{n+1} < 1$ too and even though we know that $a_{n+1} > a_n$ and $a_n < 1$, this does not always imply that $a_{n+1} < 1$. So we resort to the recursive definition of the sequence $\{a_n\}_{n \geq 1}$ once more. We are given that $a_{n+1} = \frac{3a_n}{2 + a_n}$. So, proving that $a_{n+1} < 1$ is equivalent to proving that $3a_n < 2 + a_n$, i.e. $2a_n < 2$. But that is true because by the induction hypothesis (5) (which is stronger than the earlier induction hypothesis (1)), we now know that $a_n < 1$.

So we have proved both the parts of the R.H.S. of (4) by induction on n . The crucial point is that we did this simultaneously for the two parts. If we tried to prove only the first part (which is our goal), we ran into trouble without the second part. We had to take the help of the other part. This is the magic of induction. It is sometimes easier to prove a stronger statement than the one intended, because in the inductive step of the proof of the intended statement, the extra strength of the stronger statement is needed. As a biological analogy, a male of a species cannot produce another male by itself. But working together with a female of that species, the two can produce a male and a female of the next generation of that species.

We are still not through with the proof of (i). We have to prove that $\lim_{n \rightarrow \infty} a_n = 1$. Call this limit as L . Then as $n \rightarrow \infty$, both a_n and a_{n+1} tend to L . So letting n tend to ∞ on both the sides of the given recurrence relation

$$a_{n+1} = \frac{3a_n}{2 + a_n} \quad (6)$$

we get the equation

$$L = \frac{3L}{2+L} \quad (7)$$

which reduces to a quadratic $L^2 = L$ and has $L = 0$ and $L = 1$ as solutions. Since the terms of $\{a_n\}$ are all positive and increasing, L cannot be 0. So $L = 1$.

Essentially the same argument was used in answering Q.30 of Paper 1. Both the questions are of the same spirit. That question deals with a functional relation which defines the value of a function f at a point x in terms its value at $\frac{x}{2}$. In the present question, the value of a_{n+1} is given in terms of a_n . There we wanted to find the limit of $f(x)$ as $x \rightarrow 0^+$. Here we want the limit of a_n as $n \rightarrow \infty$. In that problem we first gave a sneaky solution, which consist of using the functional equation to show that $\lim_{x \rightarrow 0^+} f(x)$ is 0 *provided* it exists. The argument above to show that $L = 1$ is also of this type. But in an MCQ, the proof of existence of the limit is not needed. But in this problem we have to give it. At the end of the answer to Q.30 of Paper 1, we gave a proof by showing that the $\epsilon - \delta$ definition of a limit is satisfied. But this is usually not so easy. A far more common practice is to relate the sequence $\{a_n\}$ to some other standard sequences such as $\frac{1}{n}$ or α^n where $|\alpha| < 1$ whose convergence is already established. To do this we first need to get hold of what is called a **closed form expression** for a_n , i.e. a formula which gives its value directly in terms of n without having to calculate a_{n-1} . (Such closed form expressions are well known for the arithmetic and geometric progressions, both of which are examples of recursively defined sequences.)

To obtain such a closed form expression, we calculate a few terms of the sequence $\{a_n\}$, guess if they form any pattern and then prove this guess by induction on n . The first four terms (all expressed in terms of a_1) are as follows.

$$\begin{aligned} a_1 &= a_1 \\ a_2 &= \frac{3a_1}{2+a_1} \\ a_3 &= \frac{3a_2}{2+a_2} = \frac{3 \cdot \frac{3a_1}{2+a_1}}{2 + \frac{3a_1}{2+a_1}} = \frac{9a_1}{4+5a_1} \end{aligned}$$

$$a_4 = \frac{3a_3}{2 + a_3} = \frac{3 \frac{9a_1}{4+5a_1}}{2 + \frac{9a_1}{4+5a_1}} = \frac{27a_1}{8 + 19a_1} \quad (8)$$

The pattern that emerges from these is

$$a_n = \frac{3^{n-1}a_1}{2^{n-1} + (3^{n-1} - 2^{n-1})a_1} \quad (9)$$

for every $n \geq 1$. Of course, we have to *prove* that this holds for every n . But that is a ritual. Art often lies in recognising a pattern than in proving it. We prove (9) by induction on n . The truth for $n = 1$ is obvious since both the sides reduce to a_1 . Now assume that (9) holds for $n = k$. We have to prove it for $n = k + 1$. This is very straightforward. We start with the given recurrence relation for $n = k$ and get

$$a_{k+1} = \frac{3a_k}{2 + a_k} \quad (10)$$

By induction hypothesis, (9) holds for $n = k$. So we get

$$\begin{aligned} a_{k+1} &= \frac{3 \frac{3^{k-1}a_1}{2^{k-1} + (3^{k-1} - 2^{k-1})a_1}}{2 + \frac{3^{k-1}a_1}{2^{k-1} + (3^{k-1} - 2^{k-1})a_1}} \\ &= \frac{3^k a_1}{2(2^{k-1} + (3^{k-1} - 2^{k-1})a_1) + 3^{k-1}a_1} \\ &= \frac{3^k a_1}{2^k + (2 \times 3^{k-1} - 2^k + 3^{k-1})a_1} \\ &= \frac{3^k a_1}{2^k + (3^k - 2^k)a_1} \end{aligned} \quad (11)$$

which means that (9) holds for $n = k + 1$. Hence, by induction, it holds for all $n \in \mathbb{N}$.

From (9), it is very easy to calculate $\lim_{n \rightarrow \infty} a_n$. We divide both the numerator and the denominator by 3^{n-1} to get

$$a_n = \frac{a_1}{(\frac{2}{3})^{n-1} + (1 - (\frac{2}{3})^{n-1})a_1} \quad (12)$$

Since $|\frac{2}{3}| < 1$, we have that $(\frac{2}{3})^{n-1} \rightarrow 0$ as $n \rightarrow \infty$. Therefore, $a_n \rightarrow \frac{a_1}{0 + (1 - 0)a_1}$ which is simply 1 since $a_1 \neq 0$.

The paper setters probably did not expect this proof which requires that we first obtain a closed form expression for a_n . In fact, the proof above nowhere uses that $\{a_n\}_{n \geq 1}$ is increasing. The very fact that (i) asks to show this indicates that the existence of the limit is to be proved from the monotonicity of the sequence $\{a_n\}_{n \geq 1}$. This is a consequence of a very basic property of real numbers, viz. its **completeness**. Indeed the simplest formulation of completeness is that every non-decreasing sequence which is bounded above by some real number is convergent, i.e. has a limit. This is listed as an axiom in Comment No.2 of Chapter 16, because its proof requires the very construction of real numbers. But once you assume this axiom, all familiar results such as the Intermediate Value Property and the Mean Value Theorems can be proved rigorously.

The very fact that (i) first asks to prove that $\{a_n\}_{n \geq 1}$ is increasing and bounded above by 1 suggests that a proof invoking completeness of real numbers is acceptable.

We now turn to (ii). Here the arguments are very similar. Again the attempt to merely prove that $a_{n+1} < a_n$ by induction on n fails. So we consider the stronger statement

$$Q(n) \equiv a_{n+1} > a_n \text{ and } a_n > 1 \quad (13)$$

and prove it by induction on n . Once this is done, completeness of the real numbers guarantees that the limit $\lim_{n \rightarrow \infty} a_n$ exists. (Here we need the dual form of the Axiom above, which states that every monotonically decreasing sequence of real numbers which is bounded below is convergent.)

Once the existence of this limit, say L is proved, its calculation is by following steps (6) and (7). The direct calculation of L based on (9) is valid regardless of whether $a_1 > 1$ or $a_1 < 1$. (It may be noted, however, that even a proof based on (9) needs completeness, because completeness is needed to show that $\alpha^n \rightarrow 0$ as $n \rightarrow \infty$. In fact even the result that $\frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$, which we take as obvious, is not all that obvious. It is called **Archimedian property** and requires completeness for its proof. See the end of Comment No. 2 of Chapter 16. It is no exaggeration to say that there is hardly any non-trivial result of calculus which does not need completeness.)

This problem is a good illustration of the things that simply cannot be tested through an MCQ. In particular, it tests if a candidate can give a proof by induction. But the induction involved here is an unusual one in the sense that it is easier to prove the conjunction of two statements (viz. the two parts of the statement $P(n)$ in (5)) than to prove either one separately. A full length question asking the candidates to obtain a closed form expression for $\{a_n\}$ and then evaluating its limit could also have been asked without splitting into (i) and (ii).

CONCLUDING REMARKS

The avowed motivation for writing this commentary was the declining quality of the JEE questions, as compared with ISI entrance test. The comparison is fair because both are multiple choice tests and assume the same educational background of the candidates. So, these concluding remarks will focus largely on Paper 1. Comments about Paper 2 questions have already been included in the commentary of each question in Paper 2. Moreover as there are only eight questions in Paper 2. So, there is not much to say about the paper as a whole. Still a few remarks will be made in the end.

Admittedly, in Paper 1 some of the ISI problems were unexciting. These include Q.6 (maximising the area of an isosceles triangle having a given base and a given inradius), Q.8 (about the limit of the ratio of the sums of two A.P.'s), Q.10 (about solving an equation involving absolute values), Q.15 (finding the length of a side of a regular 12-gon), Q.17 (about evaluating an integral as a sum of a finite series), Q.23 (about a function defined through fractional exponents), Q.26 (about a quadratic having imaginary roots) and Q.29 (about the sum of an infinite series).

Also there were many questions whose non-triviality was marred by their being posed as MCQ's. These have been pointed out in the respective comments about them. Regrettably, this includes some excellent problems such as Q.21 (about maximising the number of points in a unit disc with no two closer than the radius), Q.28 (about the sum of two integrals whose integrands are inverse functions of each other) and Q.30 (about the limit of a function satisfying a functional equation). The last one is especially a loss because problems on functional equations are rare.

There are two questions, Q.18 (about differentiability at 0 of a function of the form $|x|^\alpha f(x)$) and Q.19 (about the increasing/decreasing behaviour

of a function defined by an integral) where the validity of the options was debatable. Also in Q.4 (about finding a condition for a cubic to have three positive real roots), although one of the options is implied by the data, it is weaker than what the data implies when done by a different method.

But all this is made up by the wealth and the variety of many good questions. They include Q.1 (finding the largest n such that $n + 5$ divides $n^5 + 5$), Q.3 (determining a divisor of a given polynomial), Q.4 (finding condition for a cubic to have three positive real roots), Q.5 (finding the integral part of $(2 + \sqrt{3})^3$), Q.9 (testing differentiability of the maximum of two functions), Q.11 (about counting permutations which are monotonic on each of two complementary subsets), Q.12 (about quadratics with even positive integers as roots), Q.13 (a locus problem), Q.14 (about conormal points on a conic), Q.16 (trigonometry problem reminiscent of heights and distances problems), Q.21 (about the maximum number of points that can be put in a unit disc subject to a distance restriction), Q.24 (about the number of invertible elements in the ring $\mathbb{Z} + i\sqrt{5}\mathbb{Z}$), Q.25 (a simple problem about power sets), Q.27 (about the behaviour of the difference between powers of consecutive integers), Q.28 (about a sum of two integrals whose integrands are inverse functions of each other) and Q.30 (about a functional equation). One only wishes that some of these were asked as full length questions to truly test a candidate's thinking.

What is equally remarkable is the absence of certain topics. For example, there are no questions about differential equations. There are no questions which simply test the ability to evaluate a tricky definite integral. No questions on vectors, none on probability, no questions on matrices and determinants, no questions about finding the equation of a circle, a line or a plane etc., no questions about systems of linear equations, no questions about trigonometric equations and, last but not least, no questions about the continuity of the integral part function. Obviously, the paper setters do not feel compelled to cover every area and to ask worn out questions on it. Rather the questions seem to be selected more on the basis of the qualities needed to solve them honestly.

In comparison to Paper 1, Paper 2 was disappointing. The first problem (about the players in a tournament) is a good one. It is a standard result in graph theory and students who are familiar with it will have an advantage. Q.2 and Q.3 are both based on elementary symmetric functions of the roots of a polynomial. Such duplication should have been avoided. Q.4 about right angled triangles was a mediocre one. So were Q.5 (a geometry problem) and

Q.6 (a trigonometry problem). But as noted there even such questions have a place today. It should also be noted that if a problem is very challenging, very few candidates can do it. Such problems enhance the prestige of the test in the academic world but do not have much effect on the selection of the candidates. So, given the pathetic state wrought out today by exclusive reliance on MCQ's, it is not a bad idea to have a subjective test with only a few problems demanding medium level of intelligence but testing a candidate's ability to give logical reasoning.

Q.7 is a tricky problem on evaluating a definite integral while Q.8 is a good question on mathematical induction. Neither is suitable as an MCQ. So the paper setters have made good use of the freedom given by subjective format of a question.

The overall job of writing this commentary was more enjoyable than writing a commentary on the JEE papers. What made it more exciting was the elegant solutions to Q.3 and 4 in Paper 1 and the tricky solution to Q.7 in Paper 2 contributed by Siddhesh Naik and the elegant solutions to Q.5, Q.23 and Q.26 in Paper 1 contributed by Deepanshu Rajvanshi. Let us hope that the day is not too far when the JEE contains a subjective component again and regains its old glory so that writing a commentary on its papers becomes enjoyable again.